Import Data using Transform Maps (Spreadsheet)

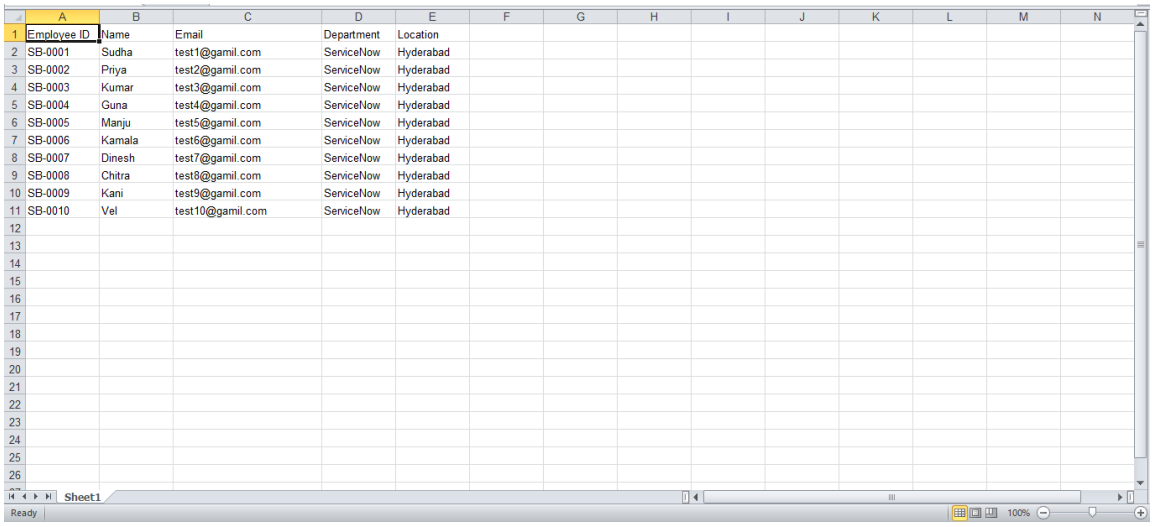
Description:

This micro project focuses on importing structured data from an external spreadsheet into the **ServiceNow** platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

Step by Step process:

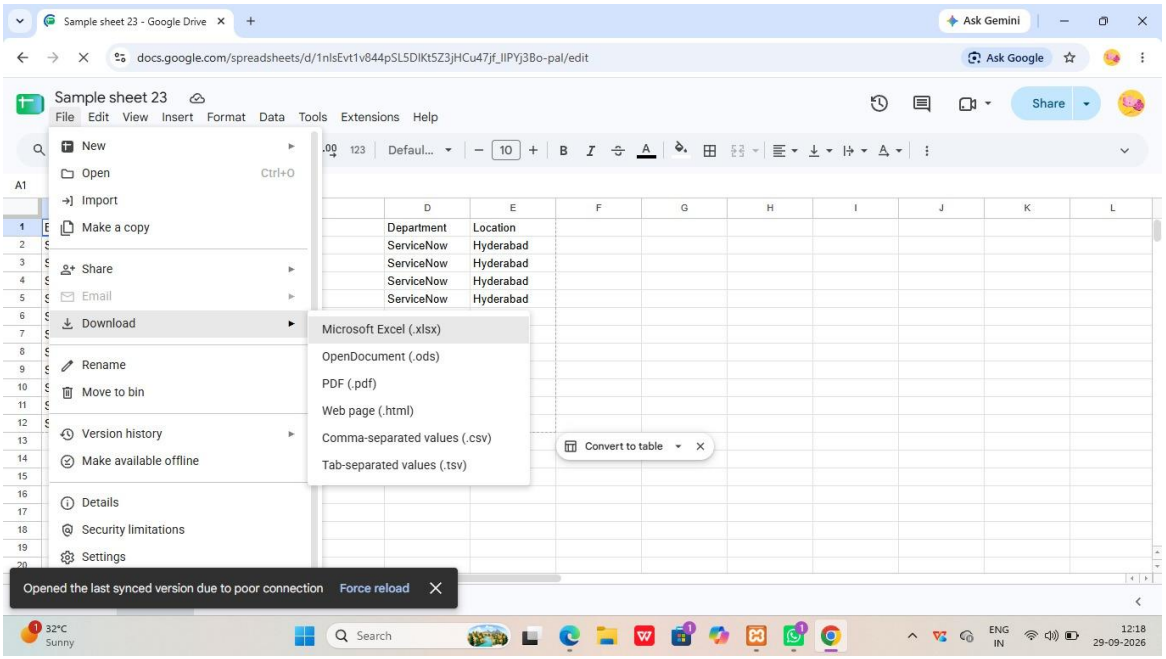
Phase 1: Prepare the sheets

1. Open New Google Spreadsheet
2. Create New Google Spreadsheet with sample data as shown below



	A	B	C	D	E
1	Employee ID	Name	Email	Department	Location
2	SB-0001	Sudha	test1@gamil.com	ServiceNow	Hyderabad
3	SB-0002	Priya	test2@gamil.com	ServiceNow	Hyderabad
4	SB-0003	Kumar	test3@gamil.com	ServiceNow	Hyderabad
5	SB-0004	Guna	test4@gamil.com	ServiceNow	Hyderabad
6	SB-0005	Manju	test5@gamil.com	ServiceNow	Hyderabad
7	SB-0006	Kamala	test6@gamil.com	ServiceNow	Hyderabad
8	SB-0007	Dinesh	test7@gamil.com	ServiceNow	Hyderabad
9	SB-0008	Chitra	test8@gamil.com	ServiceNow	Hyderabad
10	SB-0009	Kani	test9@gamil.com	ServiceNow	Hyderabad
11	SB-0010	Vel	test10@gamil.com	ServiceNow	Hyderabad

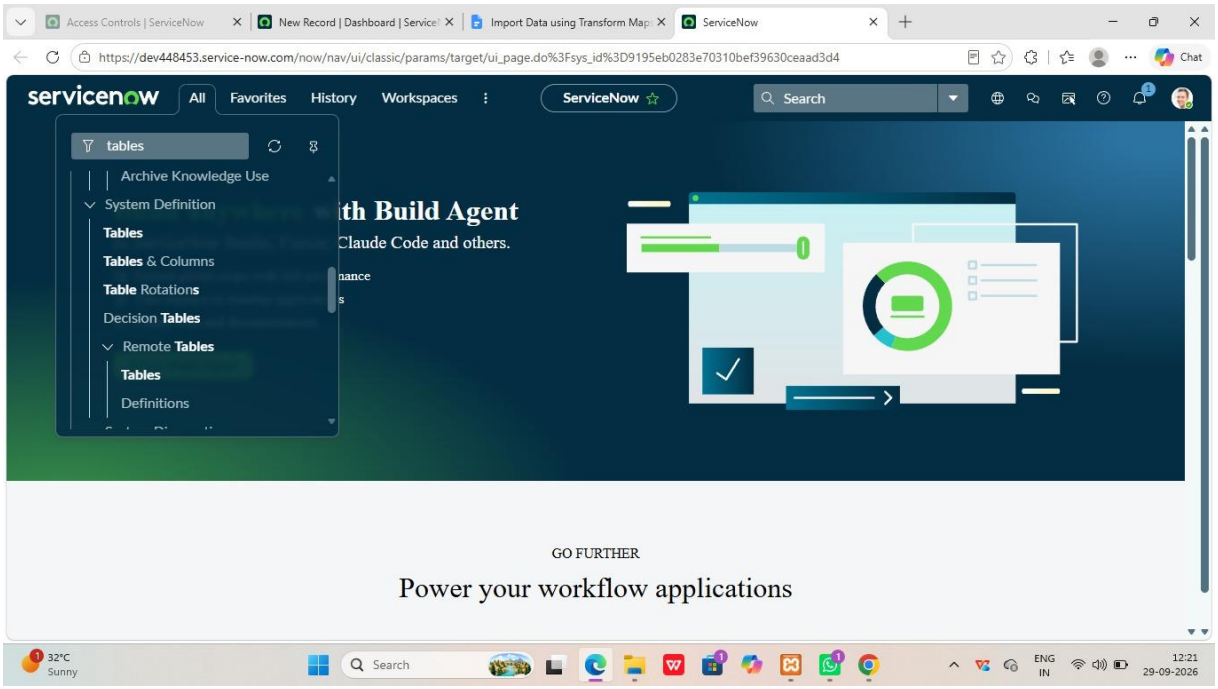
3. Download the spreadsheet as Sample Spreadsheet.xlsx format.



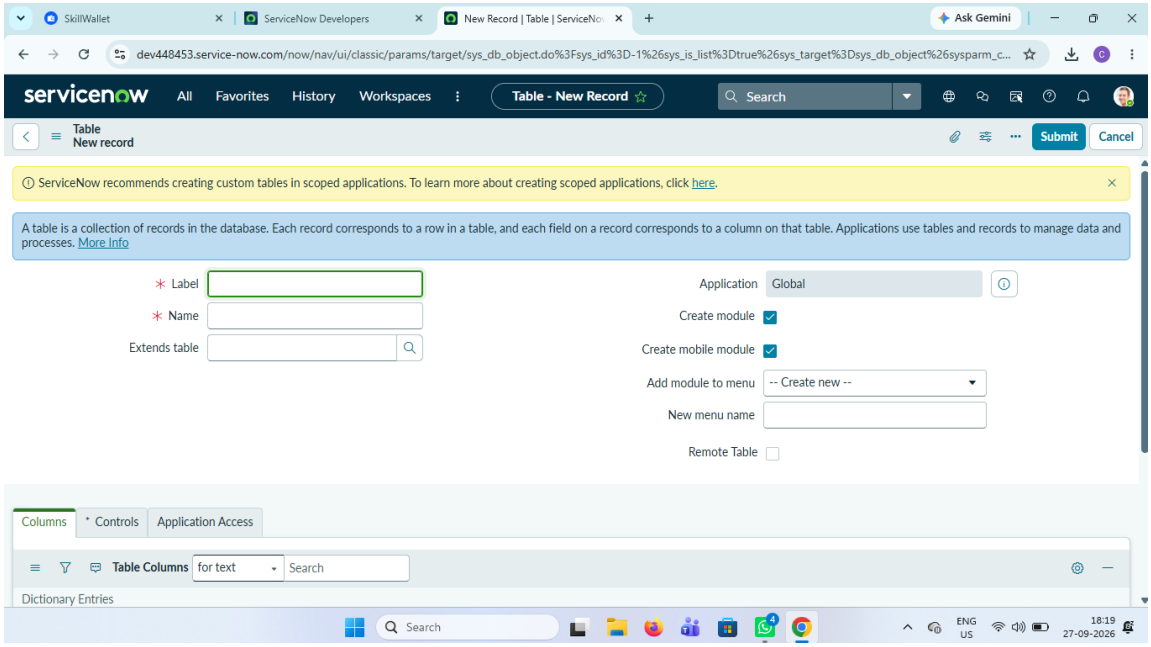
Phase 2: Create new Custom table

Navigation:

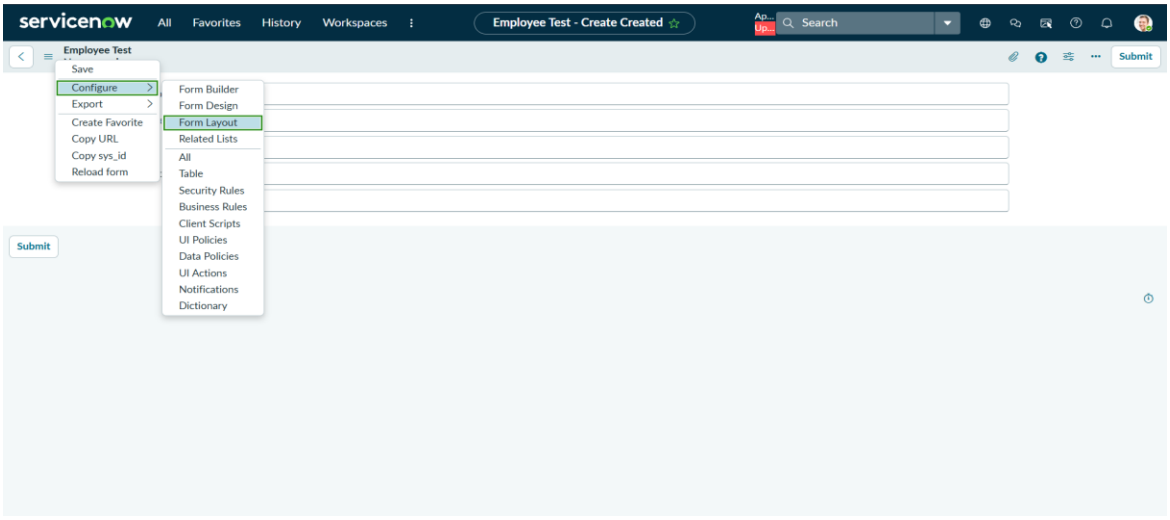
- Tables > Create New



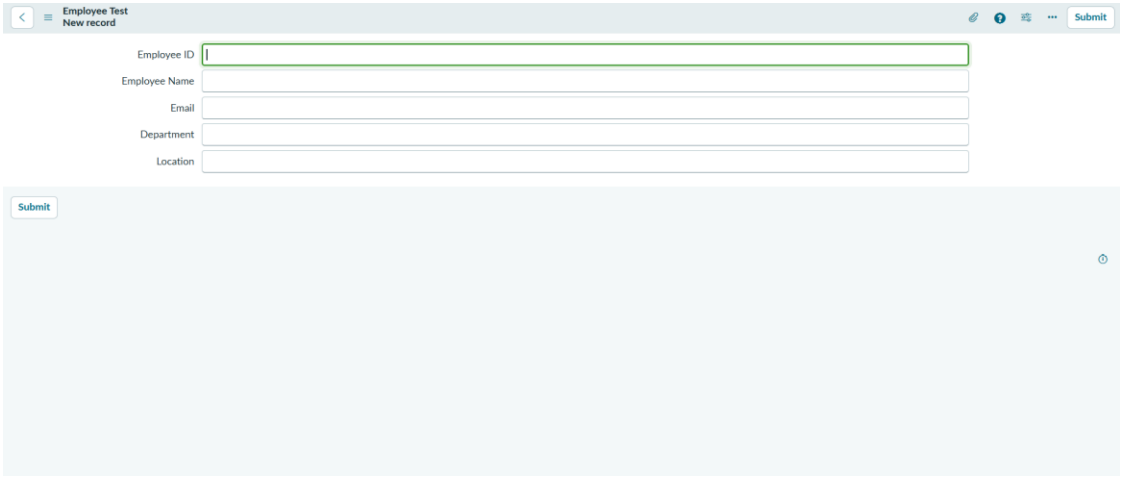
- Label: Employee Test
- Name: u_employee_test
- After creating and saving the **Employee Test** table, **scroll down** to the bottom of the table form.



- Click **Show Form** to open the table form.
- On the form, locate the **Form Context Menu / Additional Actions** option on the left side of the form header.
- Click the **Form Context Menu / Additional Actions** icon.
- Click configure , select **Form Layout**.



- The **Form Layout** page will open, where you can configure the fields displayed on the form
- After opening **Form Layout**, click **New** to create a new field.
- Enter the field details:
- **Column Label:** Employee ID
- **Type:** String
- Click **Add** to create the field.
- Repeat the same steps to create the following fields:
    - **Employee Name** – Type: String
    - **Email** – Type: String
    - **Department** – Type: String
    - **Location** – Type: String
- After creating all the required fields, click **Save**.
- Verify that the newly created fields are displayed on the **Employee Test** form.

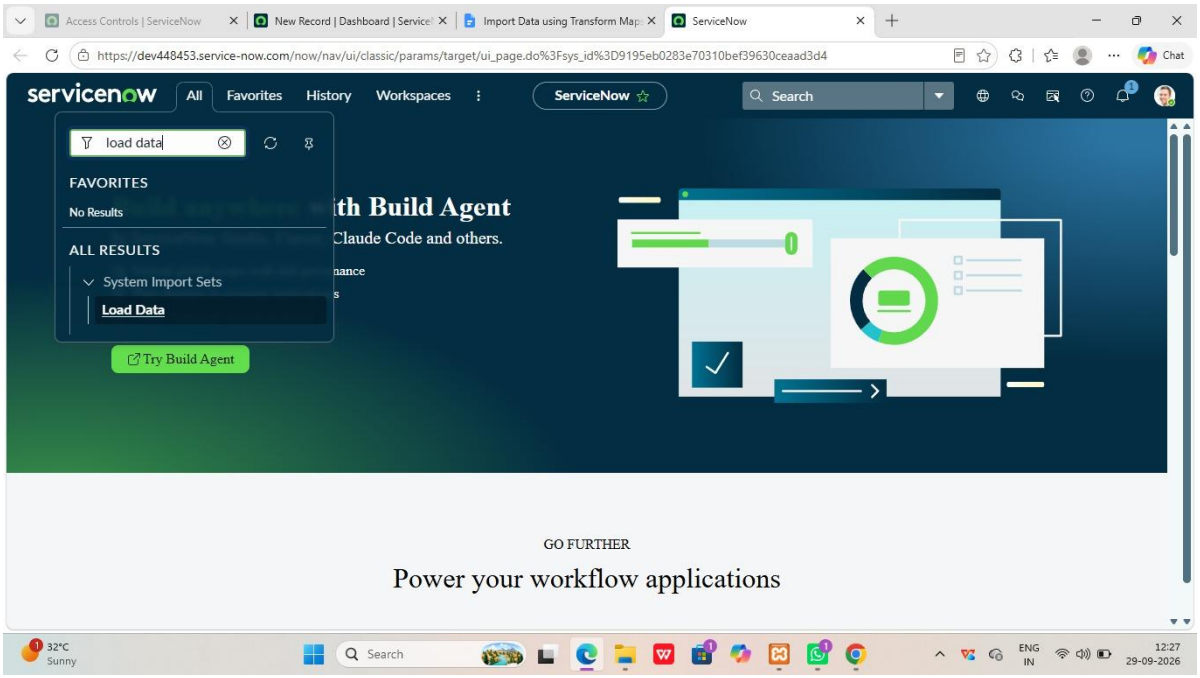


Click on Submit or Save.

Phase 3: Import Set Table

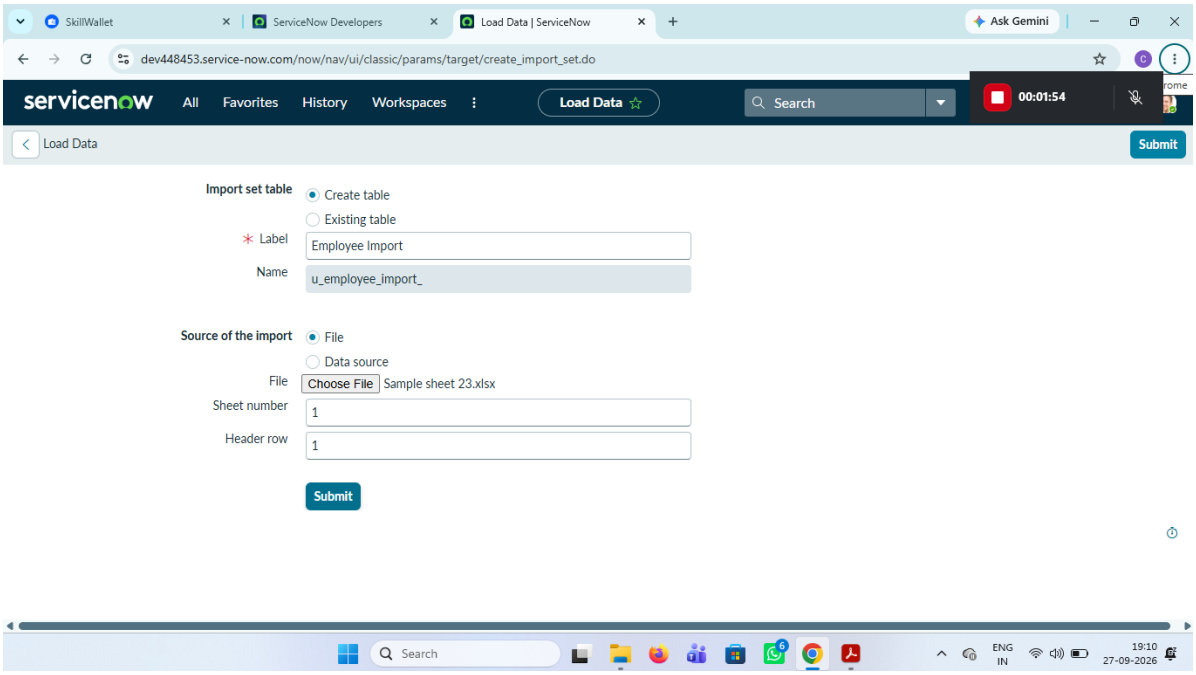
Navigation:

In Application Navigator → Load Data



Click on Load Data and create an import table
Fill details as shown below

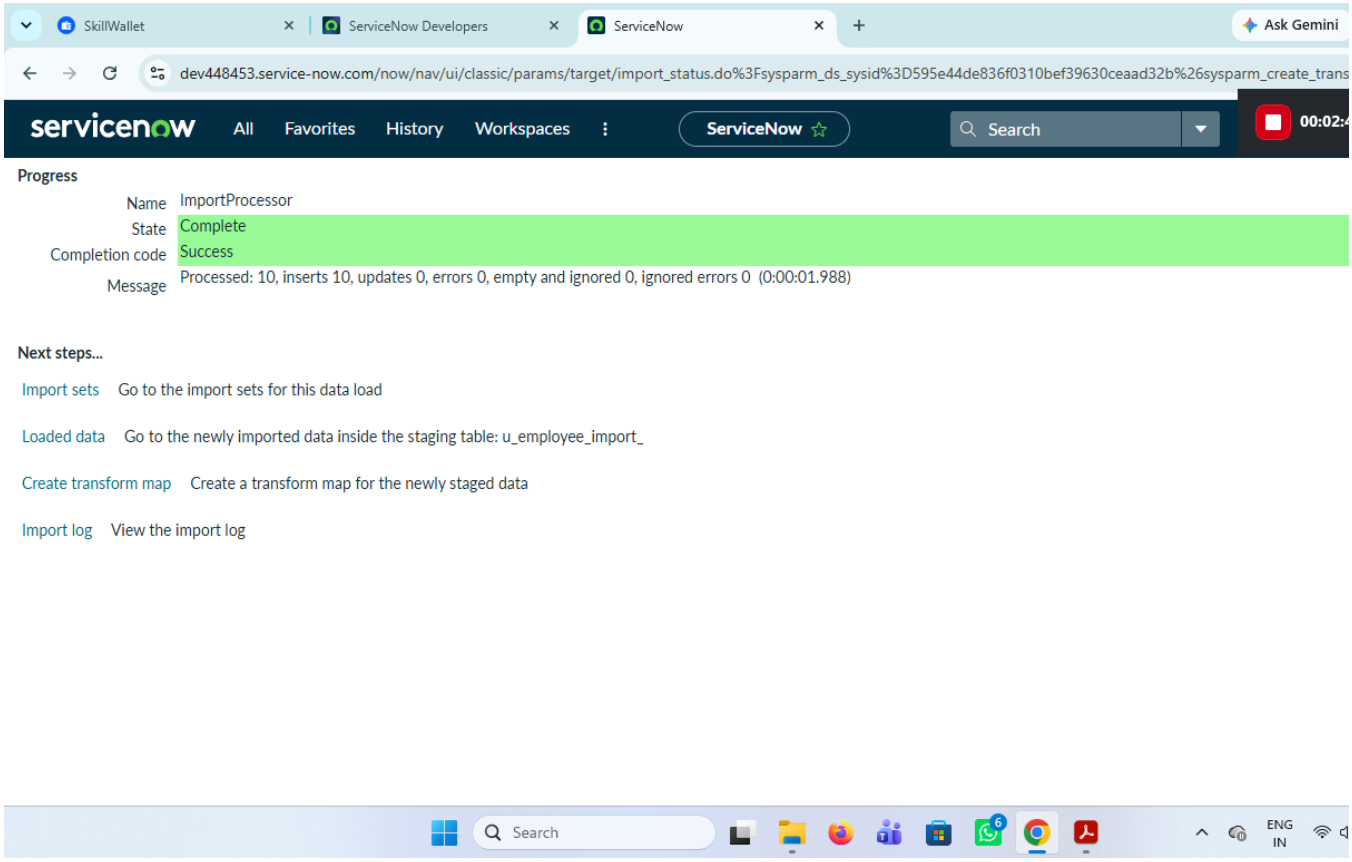
- Label: Employee Import
- Name: u_employee_import (auto populated)



- Click on Submit

Once we click on Submit, we can see the success message as shown below.

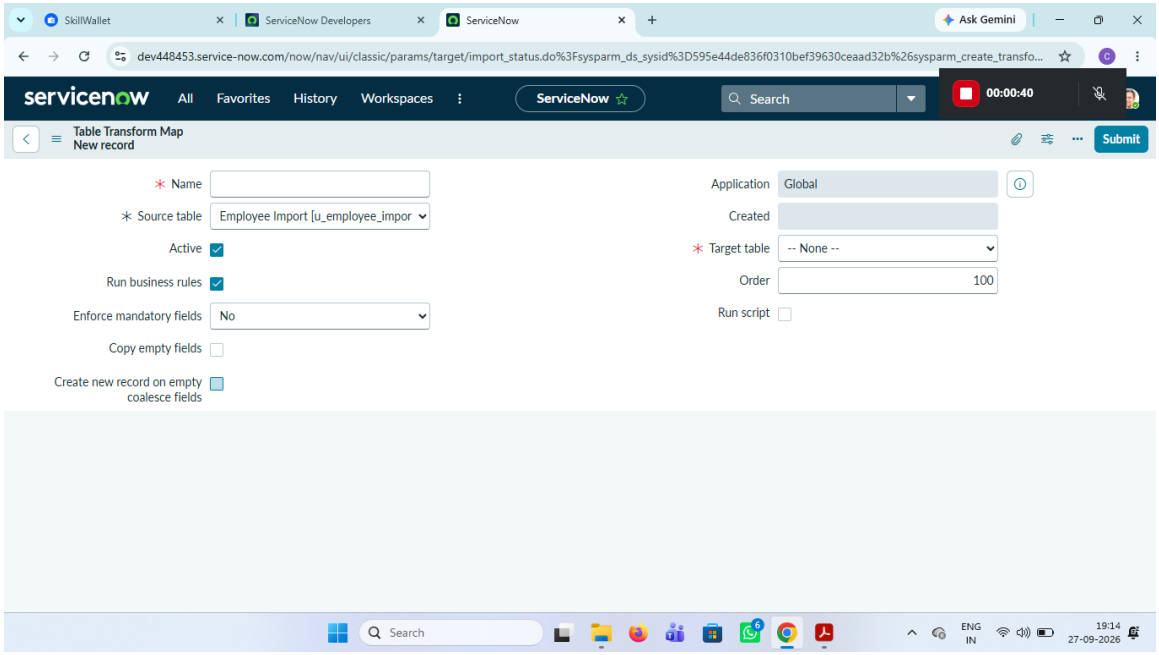
Also we need to make sure to click on Create Transform Map.



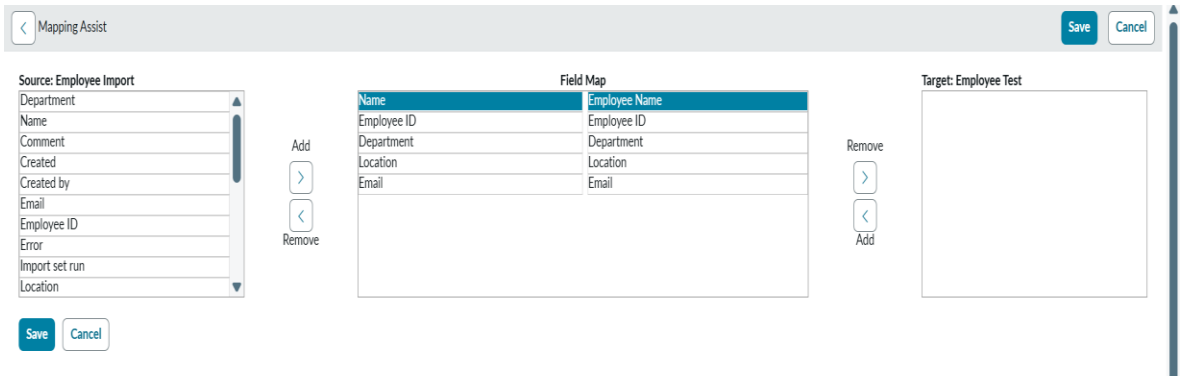
Phase 4: Create Transform Map

In this phase we need to create the Transform Map by clicking on the Create Transform Map button as shown in above screenshot.

Give inputs as shown below.
Name: Sample Spreadsheet Import
Source Table: Employee Import is auto-populated
Target Table: Employee Test - as we created earlier
Click on Auto Map Matching fields.

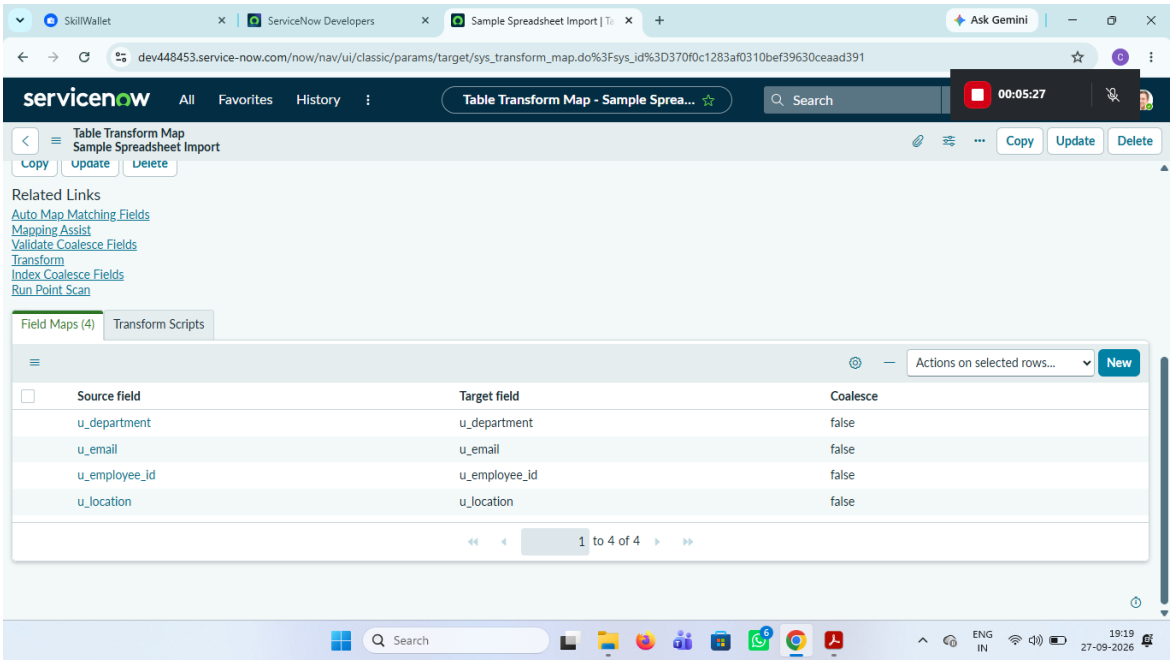


Now click on Mapping Assist and map the fields with Source & Target Table

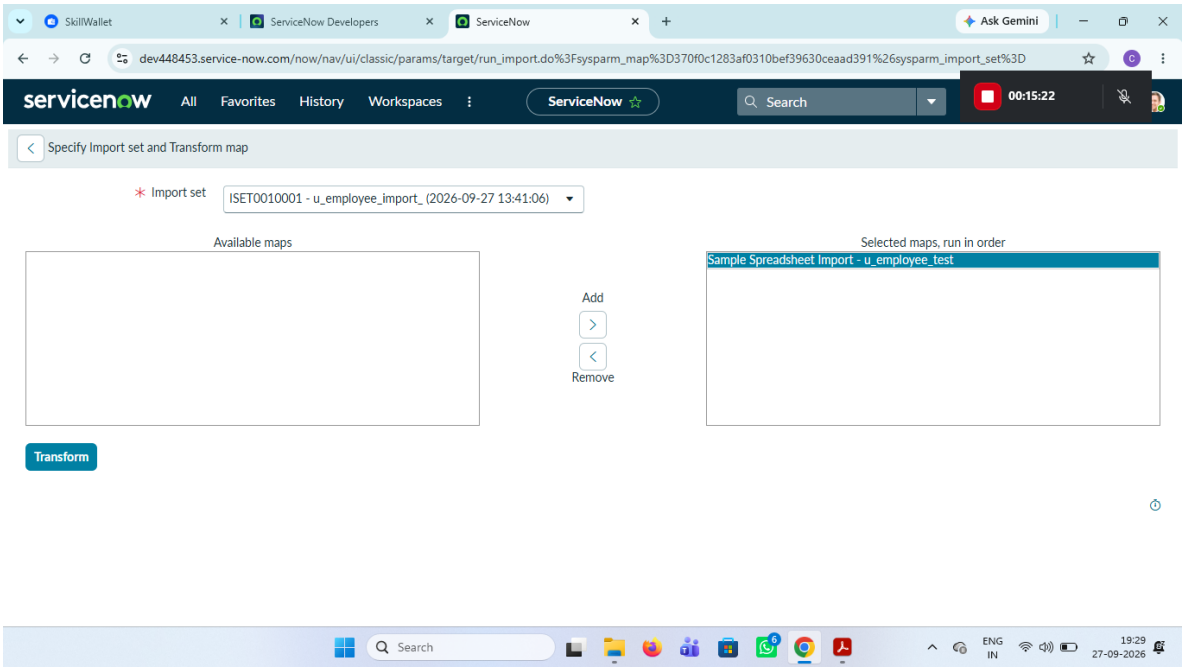


Click on **Save**

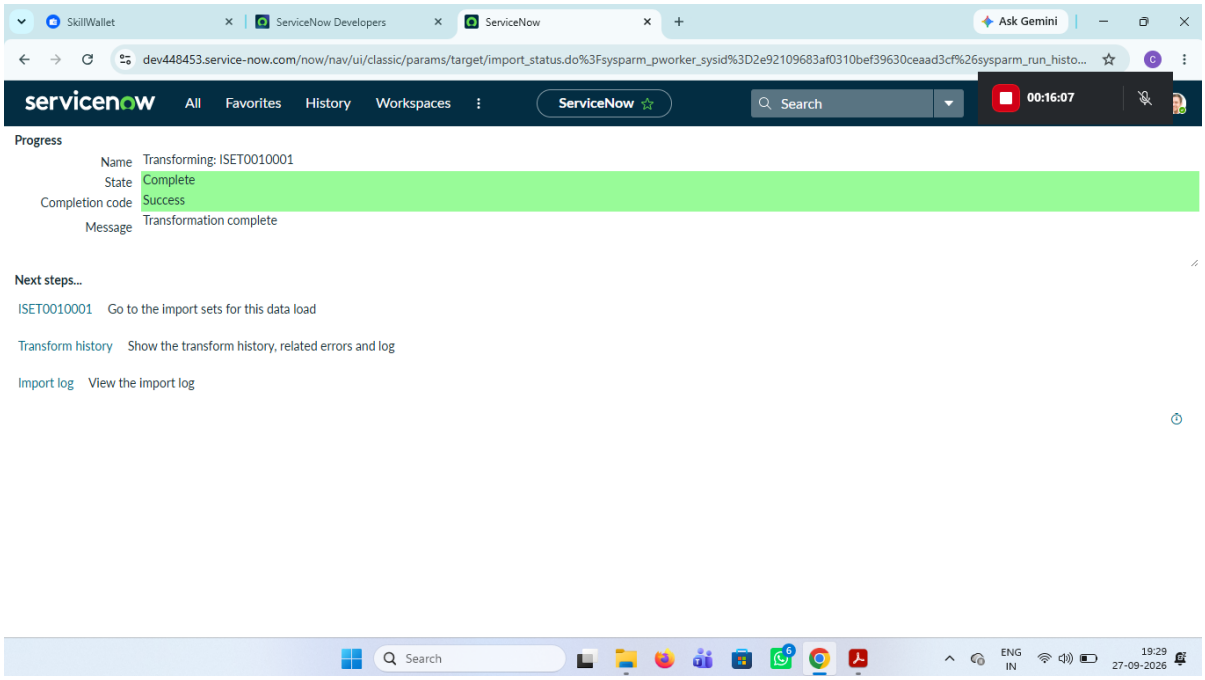
Make sure the both Source & Target table fields are mapped as shown below.



Transform Button as shown below.



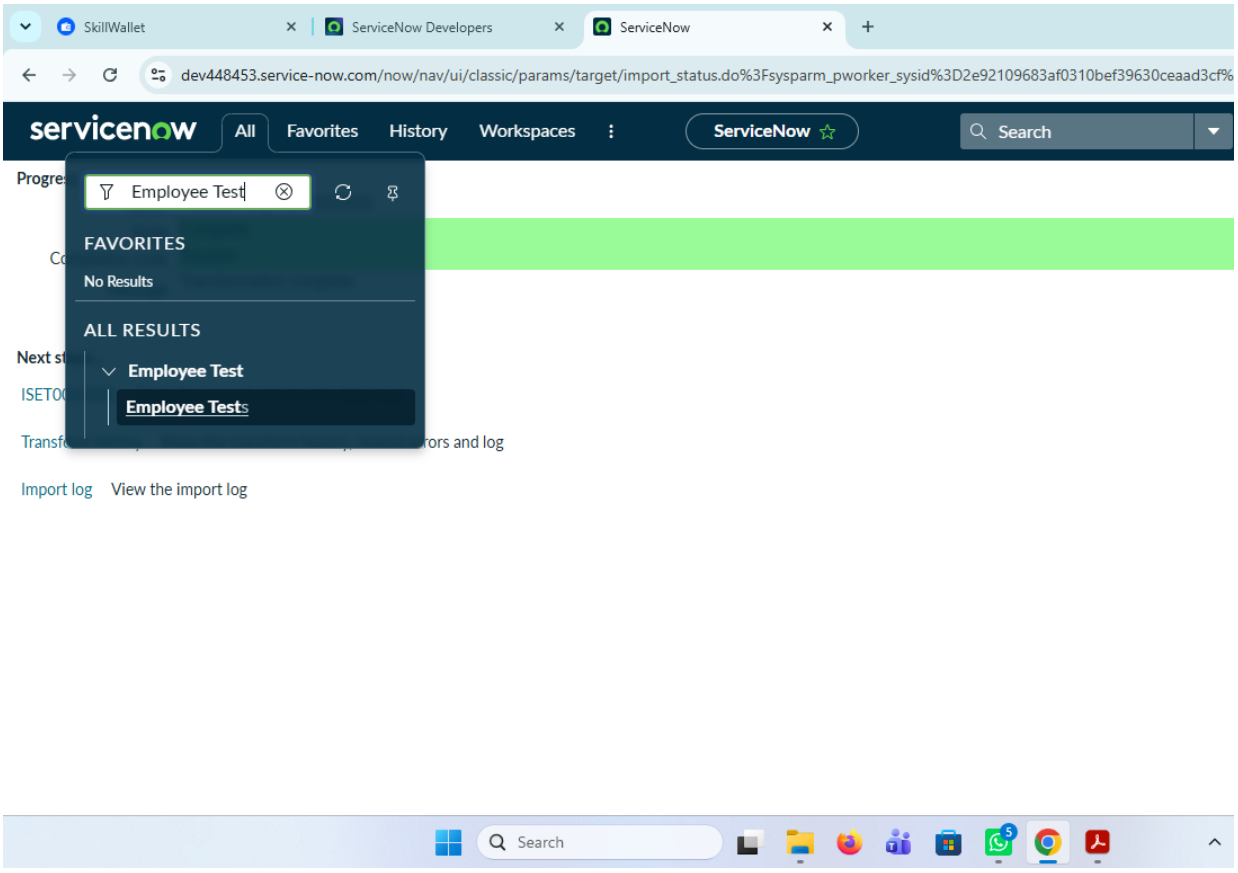
- After clicking Transform, make sure you see the success message as shown below.



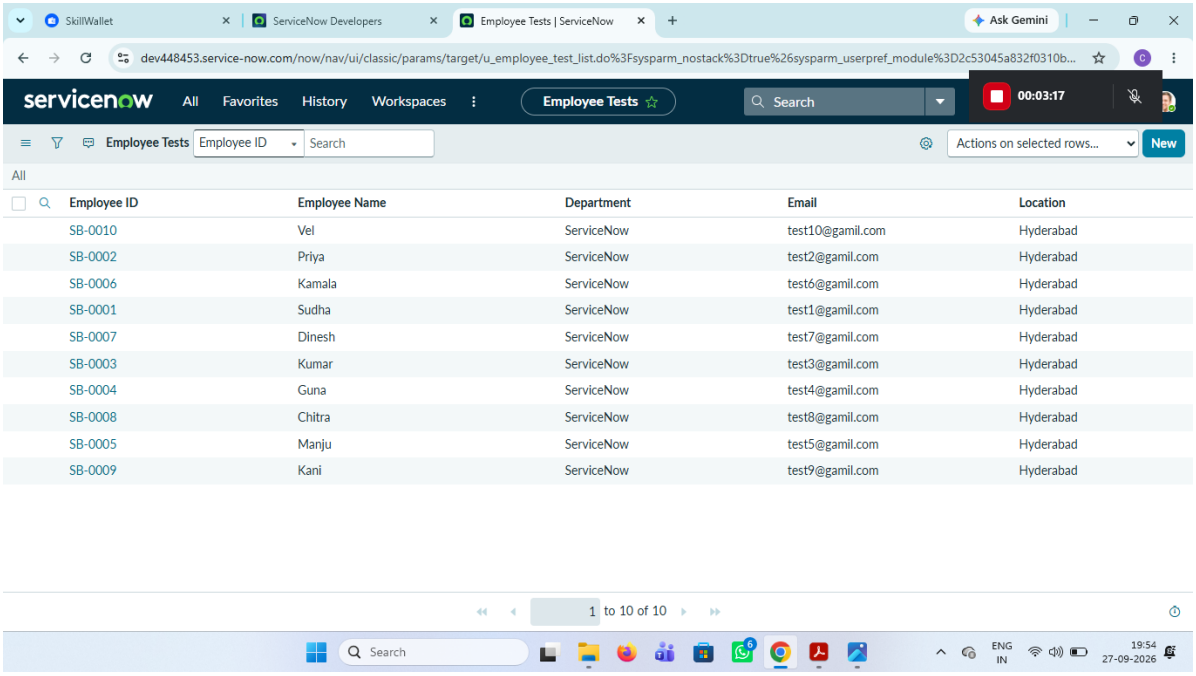
Phase 5: Transform Data & Validate

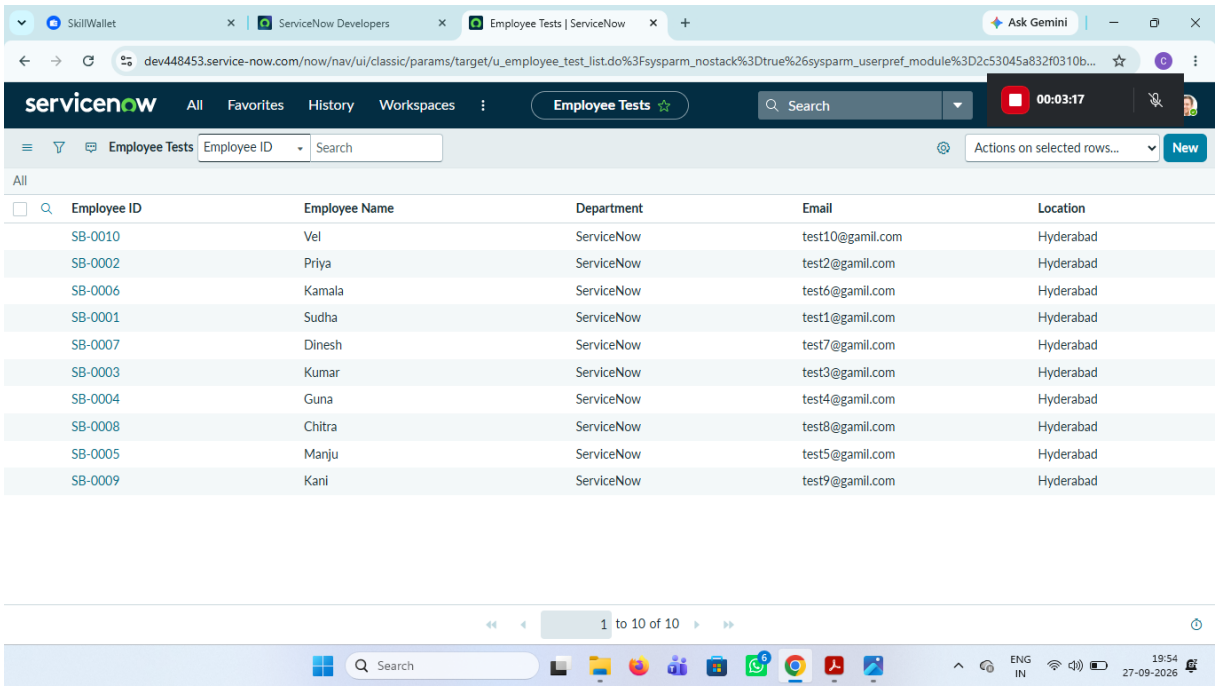
Navigation:

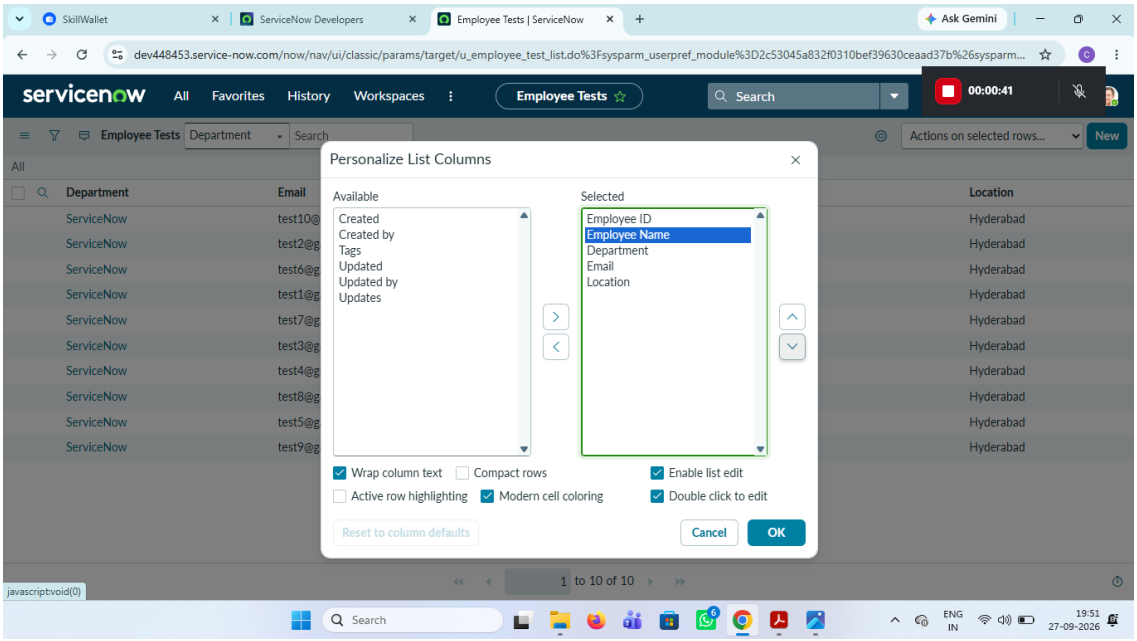
- Open Employee Test Table from application navigator as shown below.



- As our import got succeeded, all the records are created in the Employee Test Table.



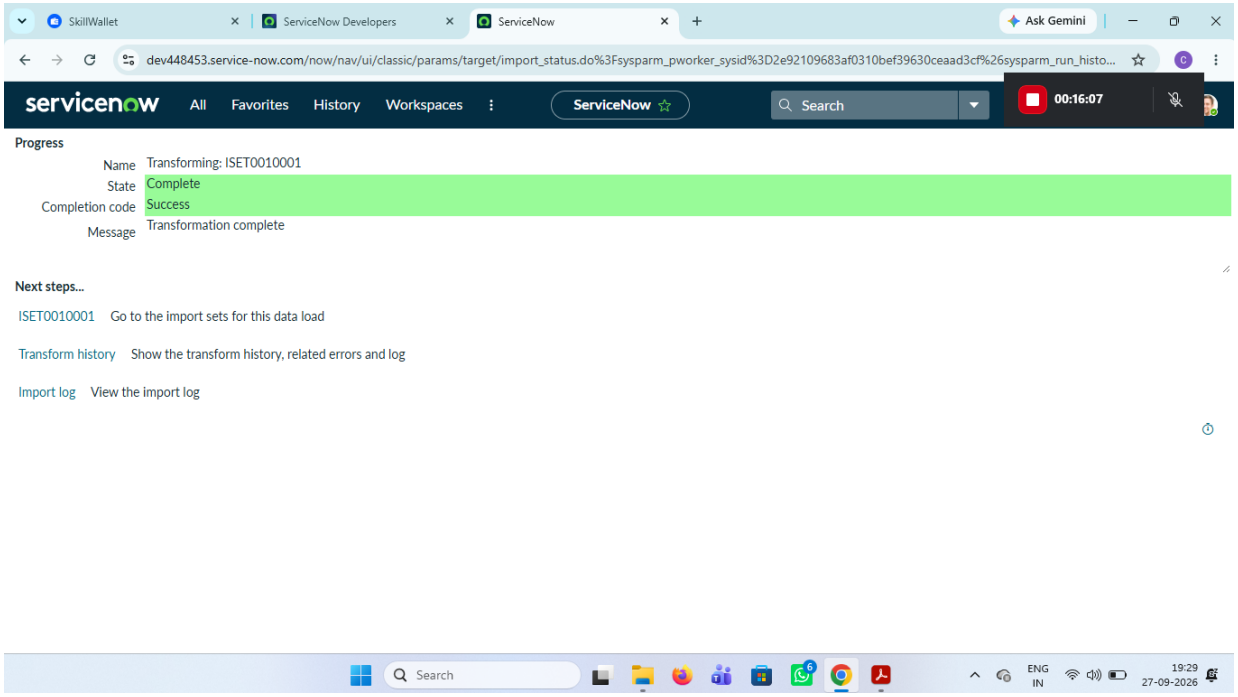




- Make sure you rearrange the fields accordingly by using Personalised List Columns

Click on OK

Final Result:
All the records are successfully imported into the ServiceNow system with Import Set, Transform Map.

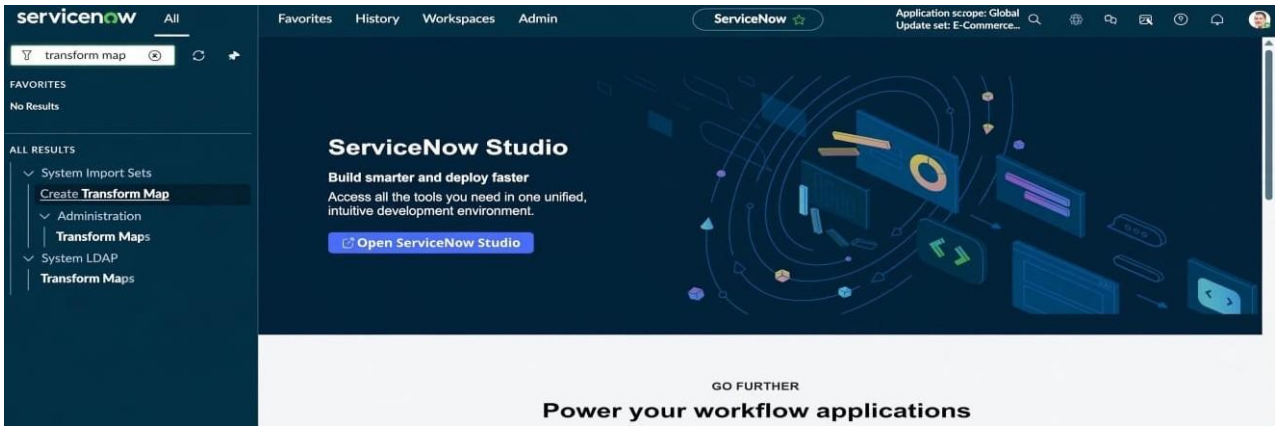


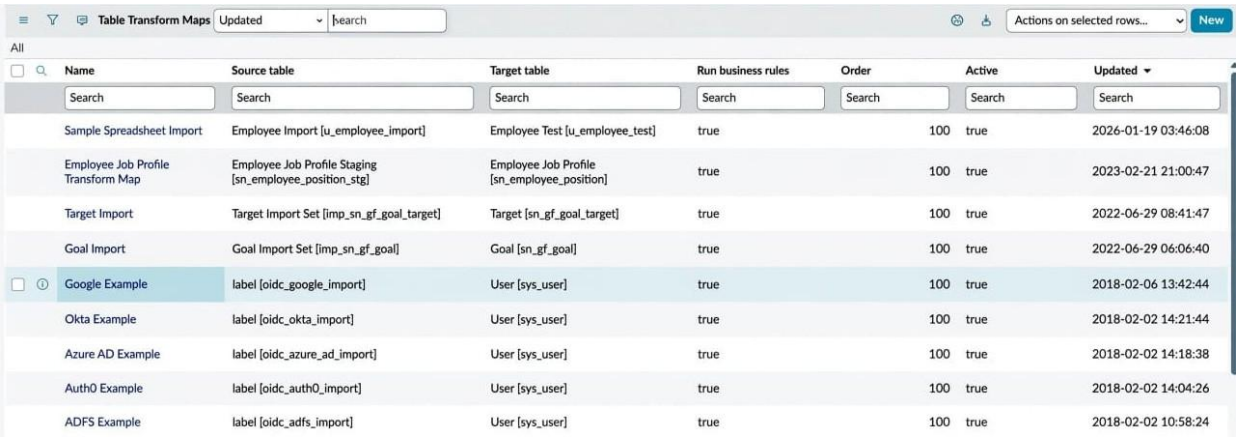
Phase 6: Enable Coalesce to Avoid Duplicate Records

- In real projects, the same employee data may be imported multiple times.
- To prevent duplicate employee records, we use **Coalesce**.
- The Coalesce will update the same records with new values if we add and the duplicate values are not inserted.

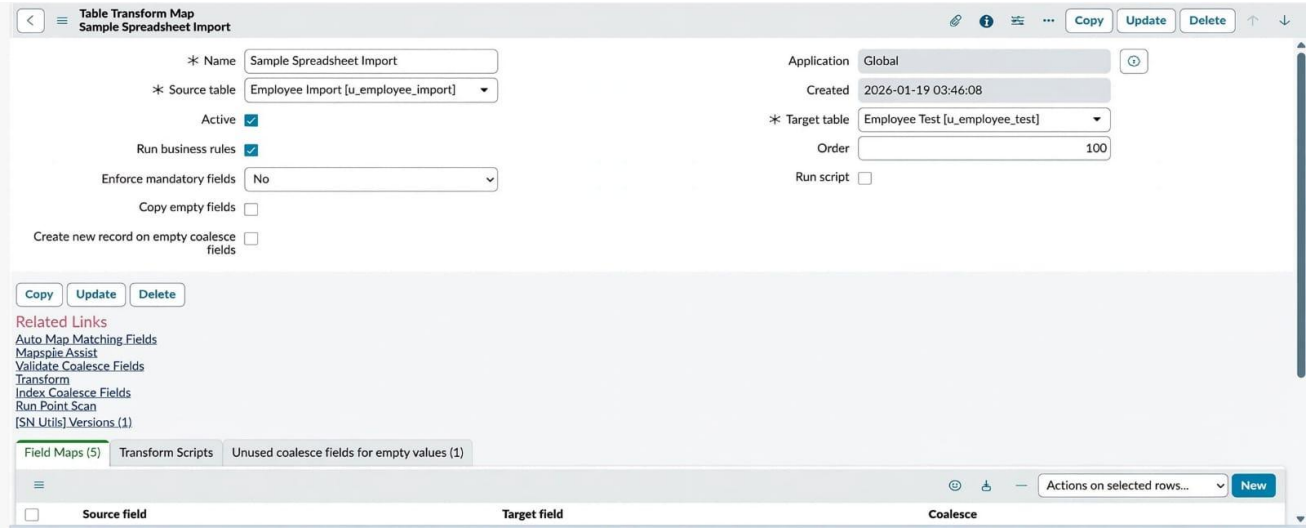
Steps:

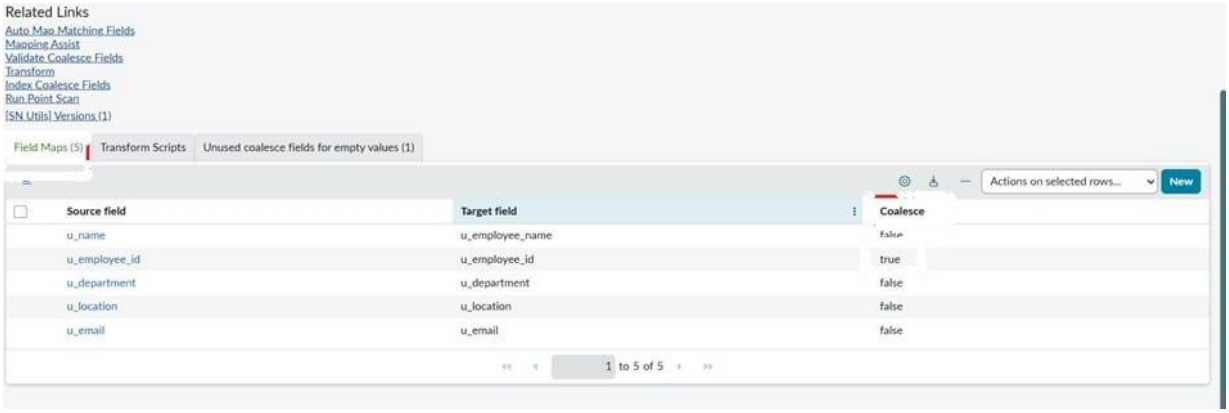
1. Open Transform map Under System LDAP On All menus.
2. Open the Created Transform map , Name is Sample Spreadsheet Import.
3. Under Related list, select the Field maps Section and Make any one field Coalesce True which appears in False.
4. Now save the form.
5. Then again add some data which is in Excel format.

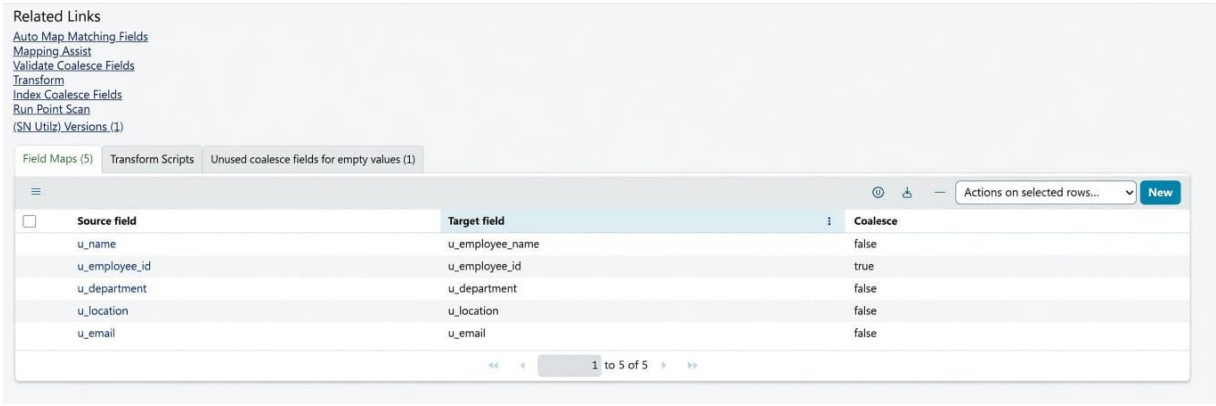




Name	Source table	Target table	Run business rules	Order	Active	Updated
Sample Spreadsheet Import	Employee Import [u_employee_import]	Employee Test [u_employee_test]	true	100	true	2026-01-19 03:46:08
Employee Job Profile Transform Map	Employee Job Profile Staging [sn_employee_position_stg]	Employee Job Profile [sn_employee_position]	true	100	true	2023-02-21 21:00:47
Target Import	Target Import Set [imp_sn_gf_goal_target]	Target [sn_gf_goal_target]	true	100	true	2022-06-29 08:41:47
Goal Import	Goal Import Set [imp_sn_gf_goal]	Goal [sn_gf_goal]	true	100	true	2022-06-29 06:06:40
Google Example	label [oidc_google_import]	User [sys_user]	true	100	true	2018-02-06 13:42:44
Okta Example	label [oidc_okta_import]	User [sys_user]	true	100	true	2018-02-02 14:21:44
Azure AD Example	label [oidc_azure_ad_import]	User [sys_user]	true	100	true	2018-02-02 14:18:38
Auth0 Example	label [oidc_auth0_import]	User [sys_user]	true	100	true	2018-02-02 14:04:26
ADFS Example	label [oidc_adfs_import]	User [sys_user]	true	100	true	2018-02-02 10:58:24



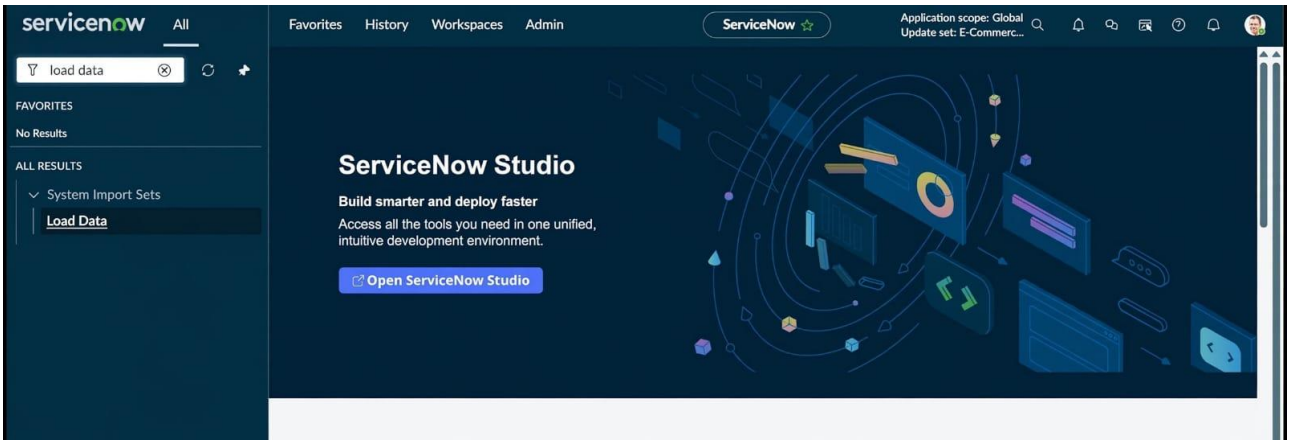


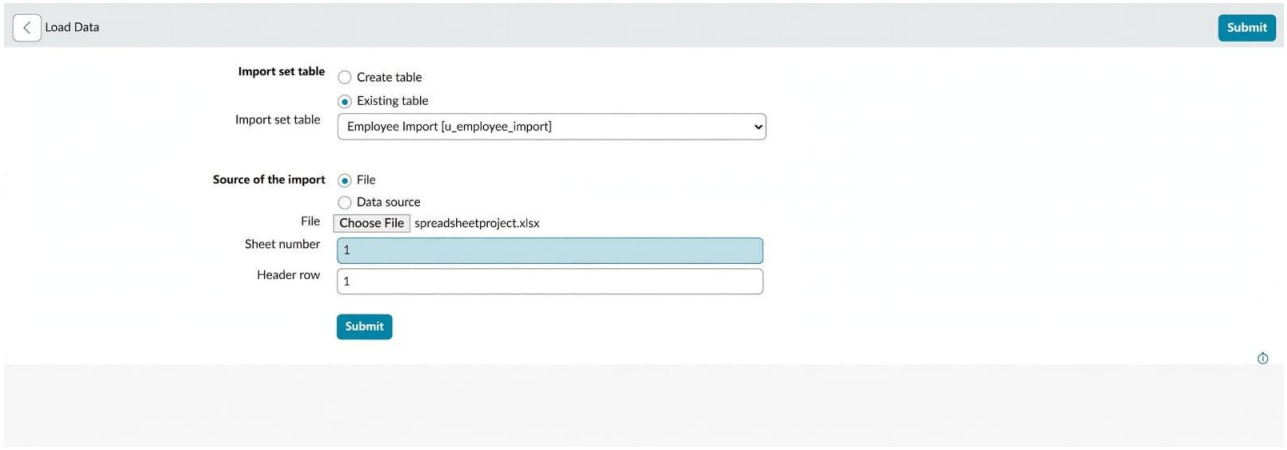


PHASE 7 : Inserting New Data In Excel Format

Steps :

- Navigate to All menus.
- Search for Load data Under System Import set Open Load Data.
- Fill the details as
- Import set table as select Existing table {Employee Import}.
- Source of Import is Select File and choose the file from Files.
- Sheet number : 1, Header row : 1.
- Now Click submit.

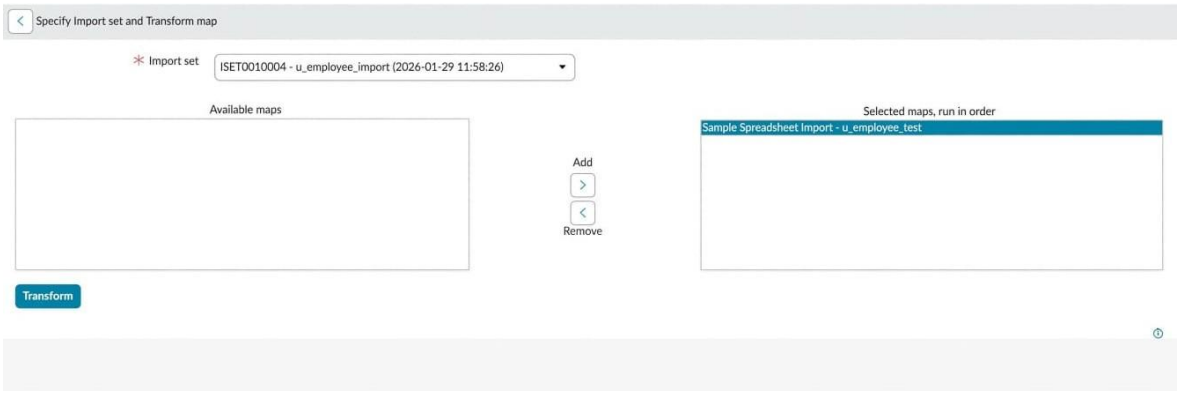




- Newly Created Excel Data is this.
- It is the previously added data.

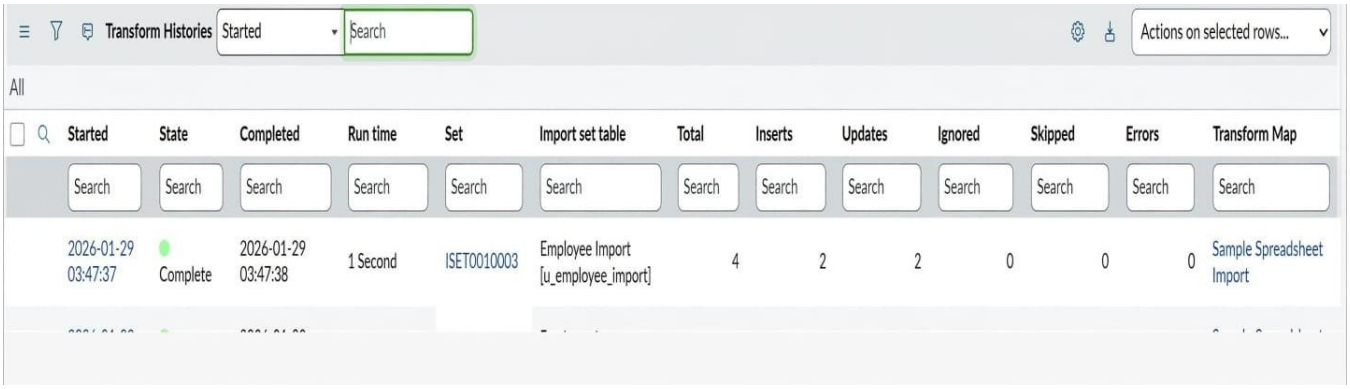


- Now Click the Run transform Under the next steps.
  - Then it opens Specify Import set and transform map which is previously selected transform map then Click Transform button.



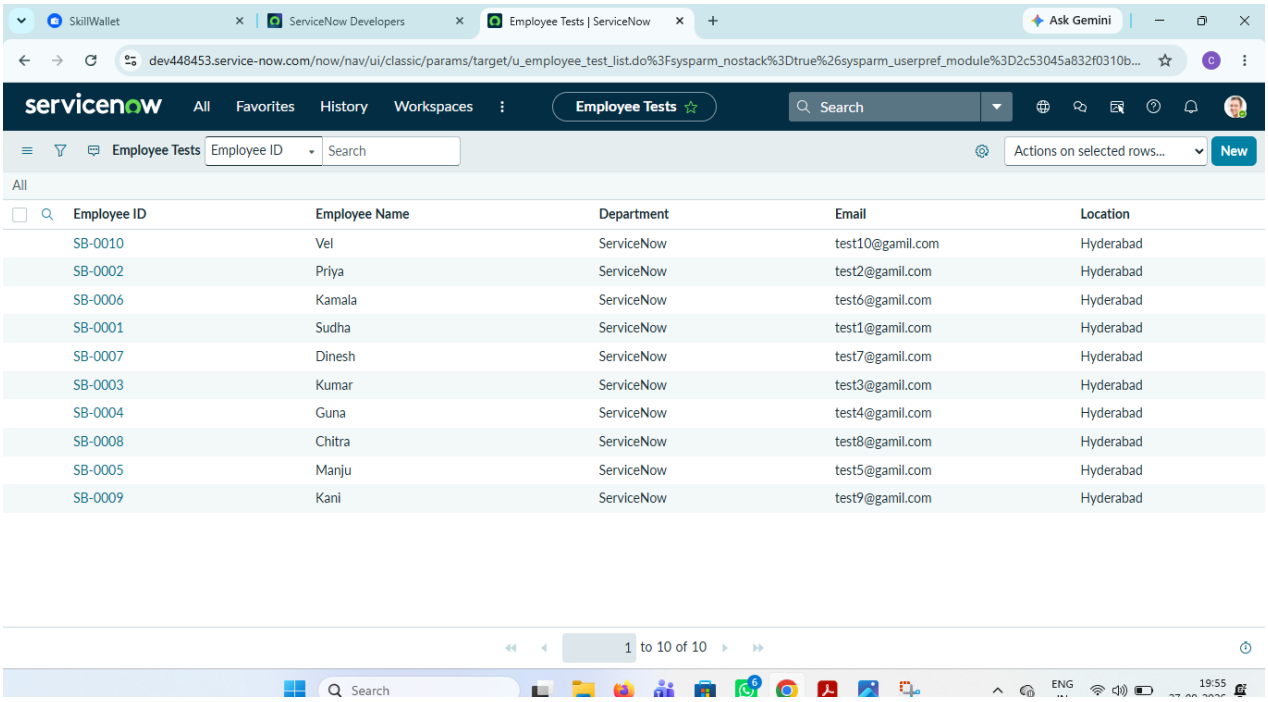


  - Now the State is completed and Success.
  - Now again click on the Transform history then it shows the newly added data and its status of Inserted and Skipped values out the given data.

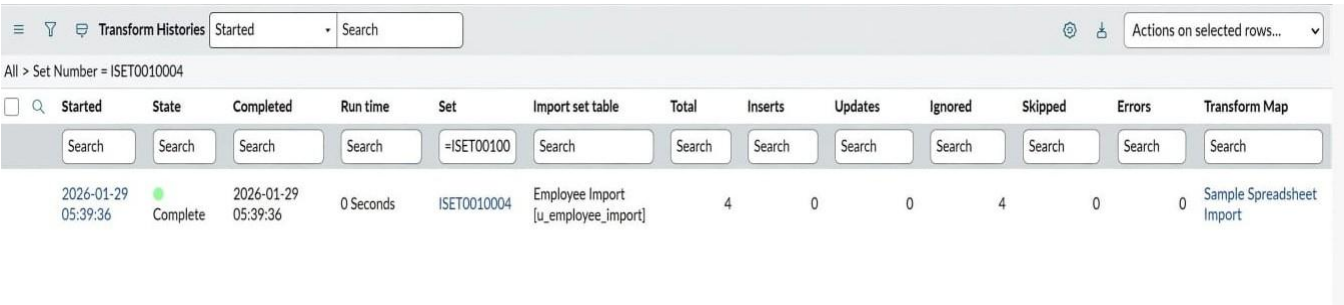


- From the above figure : total 4 rows are uploaded out of 2 are Inserted that are two new ID's.
- And 2 rows are updates that are Same ID's with different Name and different Email that we changed.

- Then Open the Employee Tests table and see the rows updated and inserted.



- Now Repeat the Phase 7 with the same steps and same Excel sheet and Submit to check how many rows are updated and Inserted with the same data.
- Then Open the Transform history to check.



- Here we can see the total 4 are uploaded.
- On that 0 rows are Inserted and updates 0 and Ignored 4 rows.
- That same sheet we uploaded to the same transform map we created.

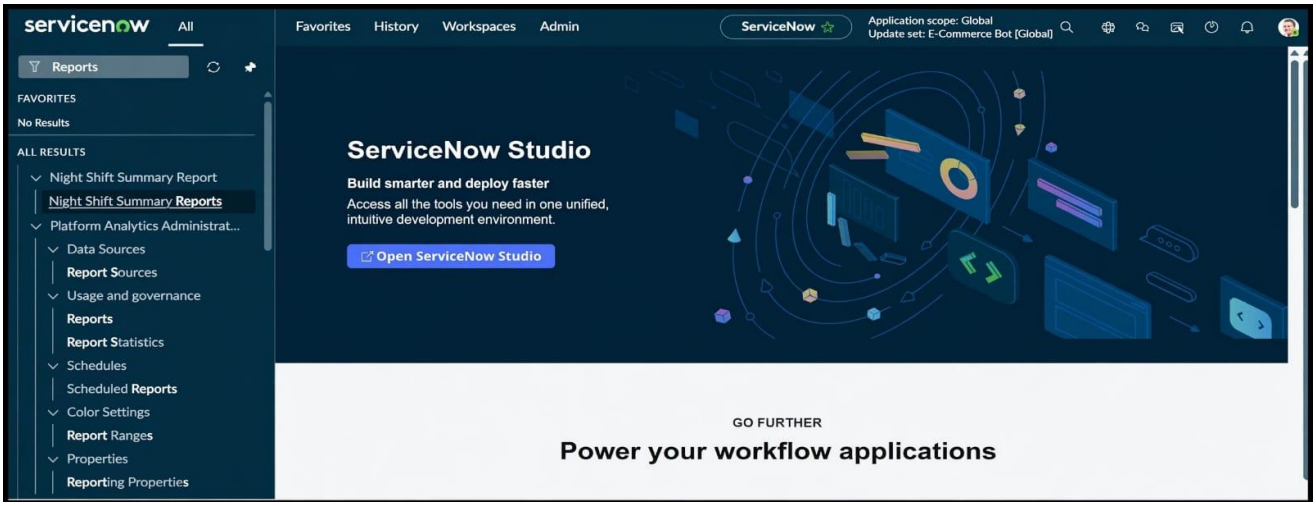
PHASE 8 : Create Reports on Employee Test Table

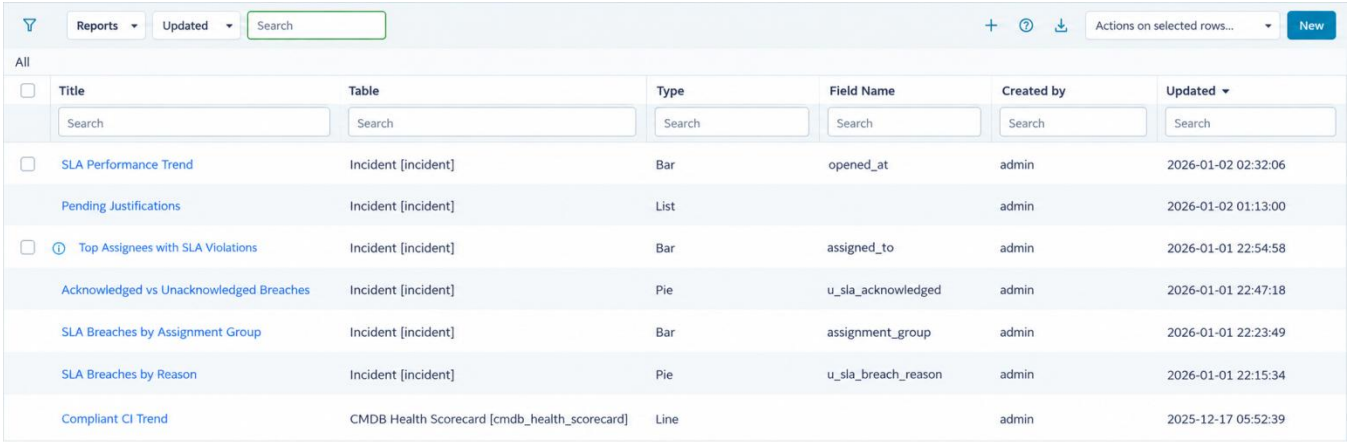
Navigation Steps :

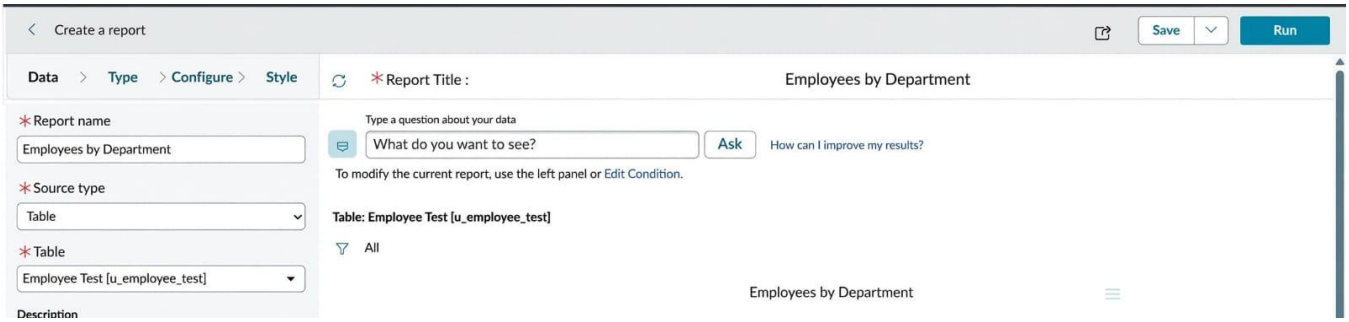
- Open All menus, Search for **Reports**.
- Under **Usage and governance** Select Reports.
- Click the **New** button to create a new report.
- Create **3 Report**s as follows.

Report 1: Employees by Department

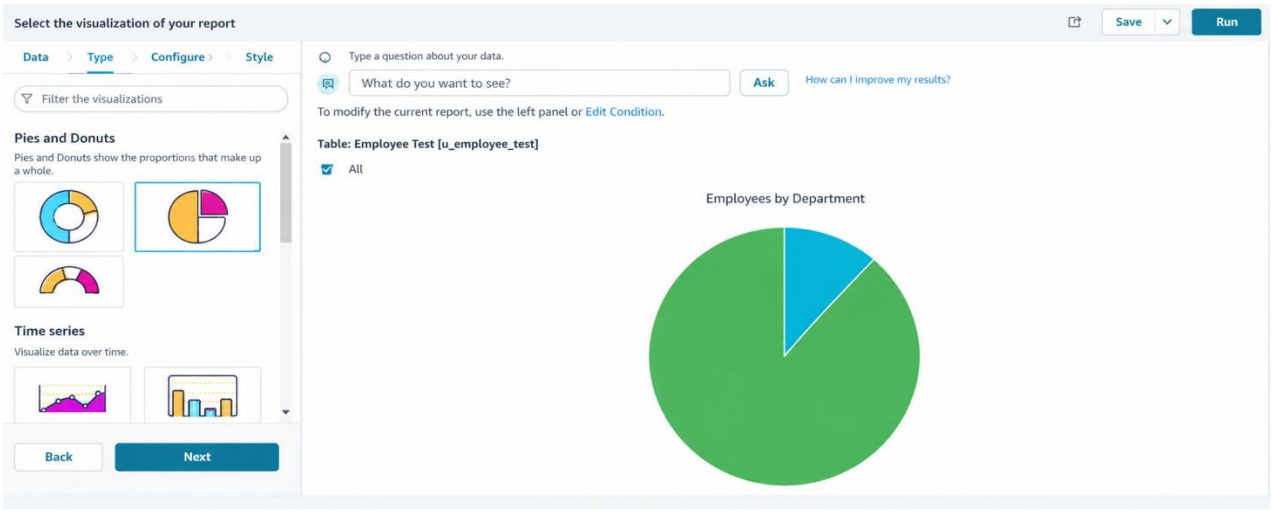
- In Data
- Report name : **Employees by Department**.
- Source Type : **Table**.
- Table : **Employee Test**, Now click the Next button.



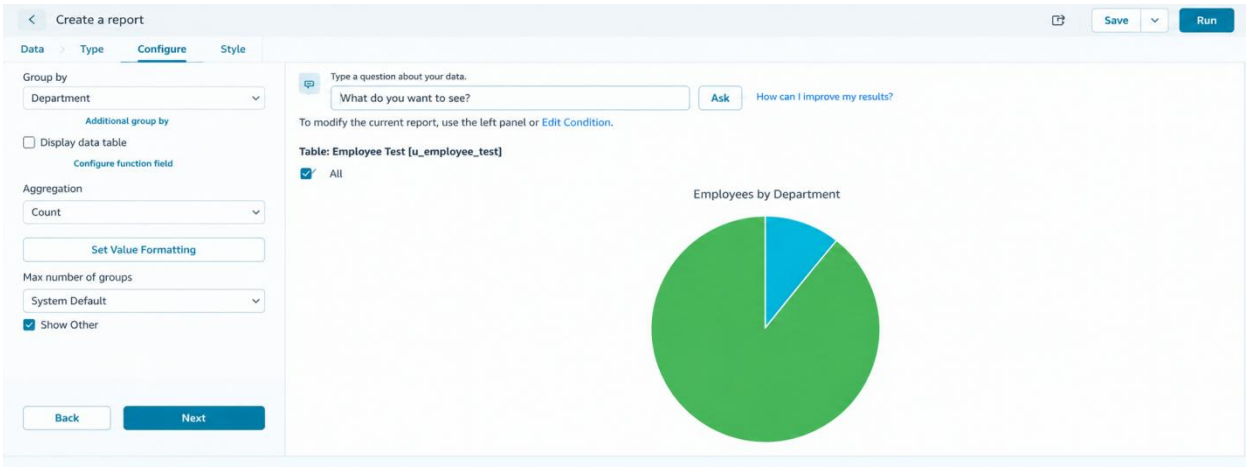




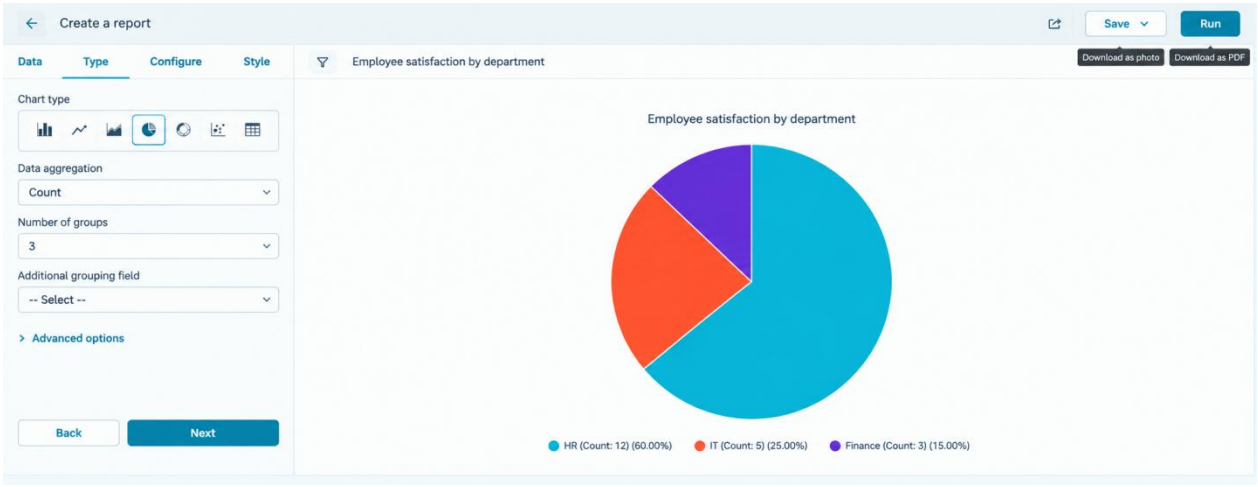
- Now after Clicking next in type Select **Pie chart** .
- And click the **next** button.



- Now In Configure Select Group by **Department** Field.
- In the Aggregation Field Select **Count.**
- Then Click **next**.



- Now In Style Leave the Chart color by **default.**
- And Click the **Run** button to Execute the given data Report.
- And Click **save** button to Save the Report.



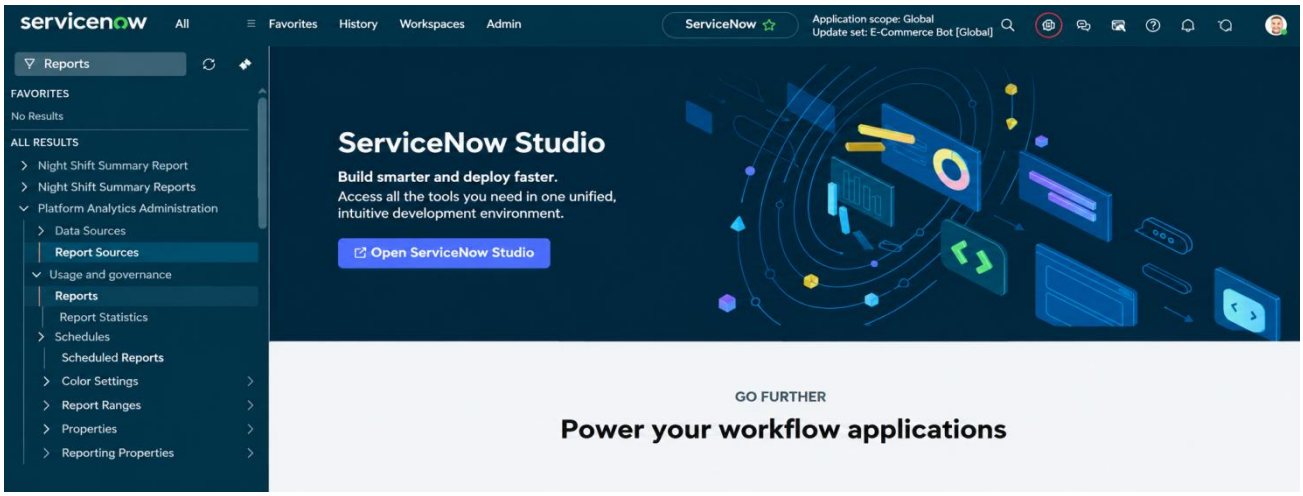
Report 2: Employees by Location

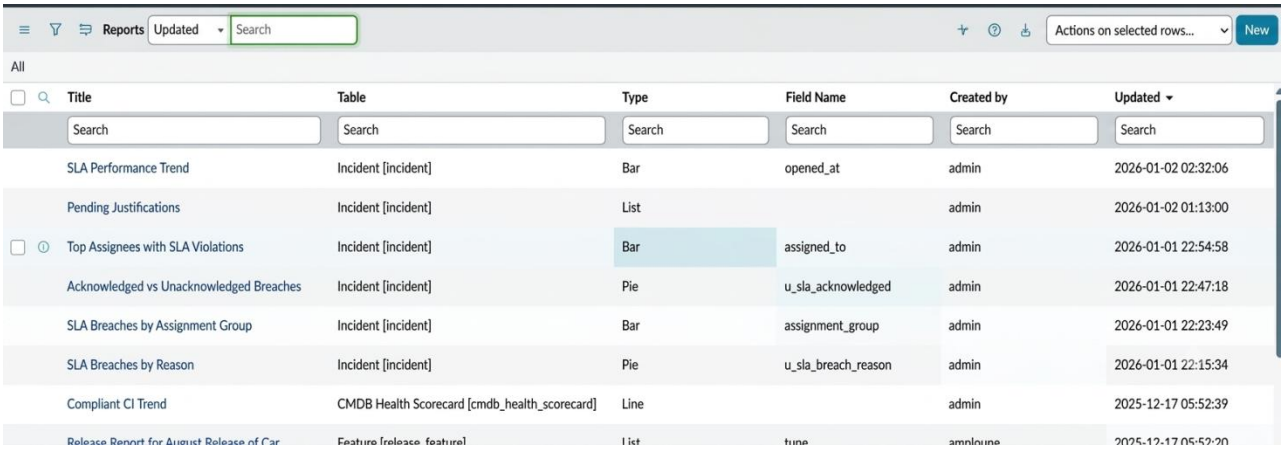
Configuration

- Table: Employee Test
- Type: **Bar Chart**
- Aggregation: **Count**
- Group By: **Location**

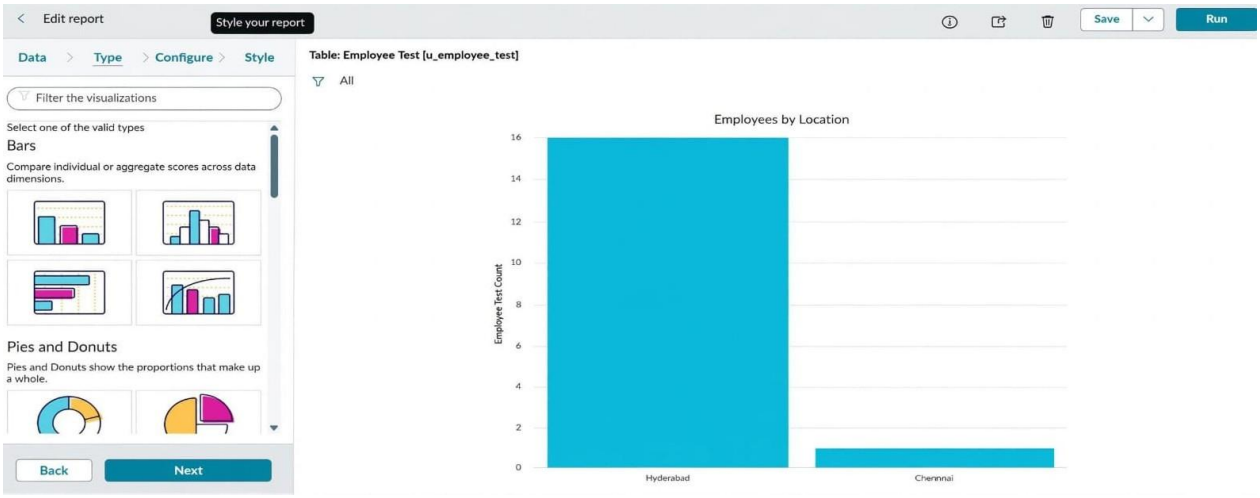
Steps :

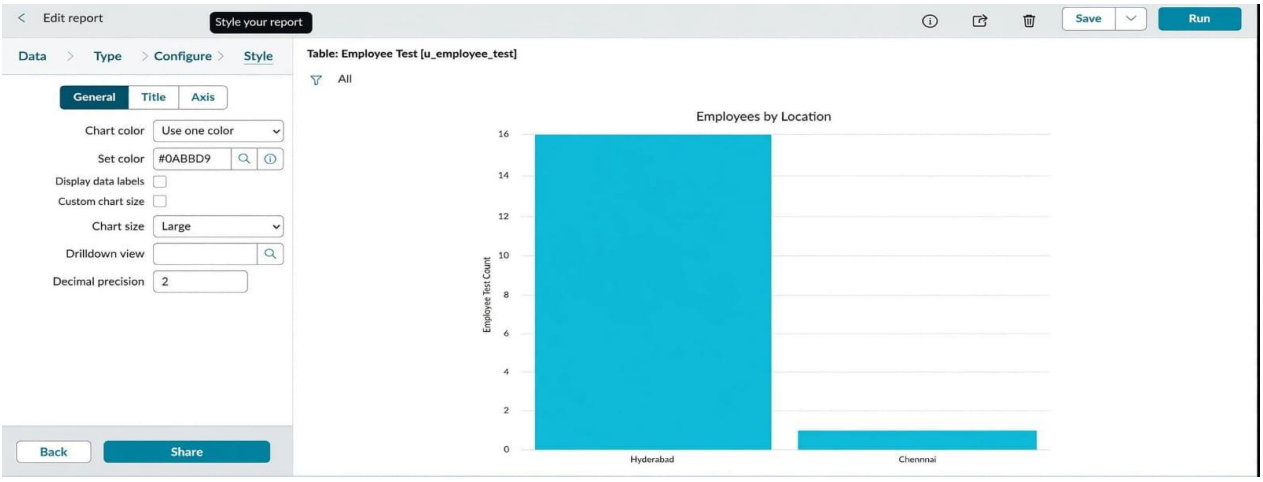
1. Create a New Report.
2. Report name : **Employees by Location.**
3. Table Select : **Employee Test**.
4. Select Type : **Bar Chart.**
5. In Configure Select Group by Field : **Location.**
6. And Aggregation is **Count**.
7. Now Click **Run** to Execute the Report and Click the **Save** to save the Report.






Edit report
Save
Run
Data
Type
Configure
Style
Report name
Employees by Location
Source type
Table
Table
Employee Test [u_employee_test]
Description
There is no description for this table. To add a description, please contact your admin.
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Count
16
14
12
10
8
6
4
2
0
Hyderabad
Chennai

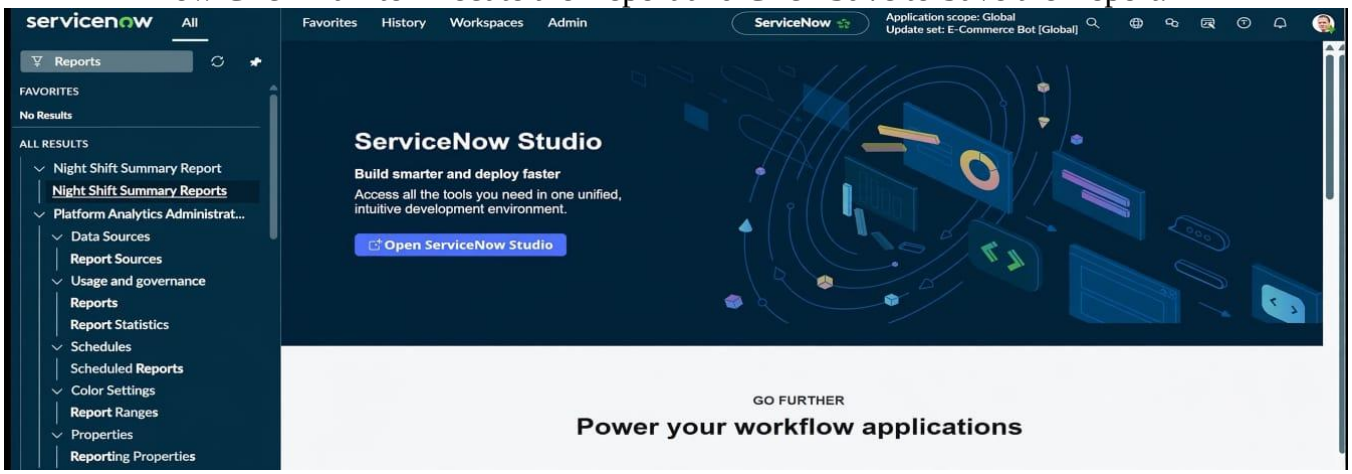

Edit report
Style your report
Save
Run
Data
Type
Configure
Style
Filter the visualizations
Select one of the valid types
Bars
Compare individual or aggregate scores across data dimensions.
Pies and Donuts
Pies and Donuts show the proportions that make up a whole.
Back
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Test Count
16
14
12
10
8
6
4
2
0
Hyderabad
Chennnai

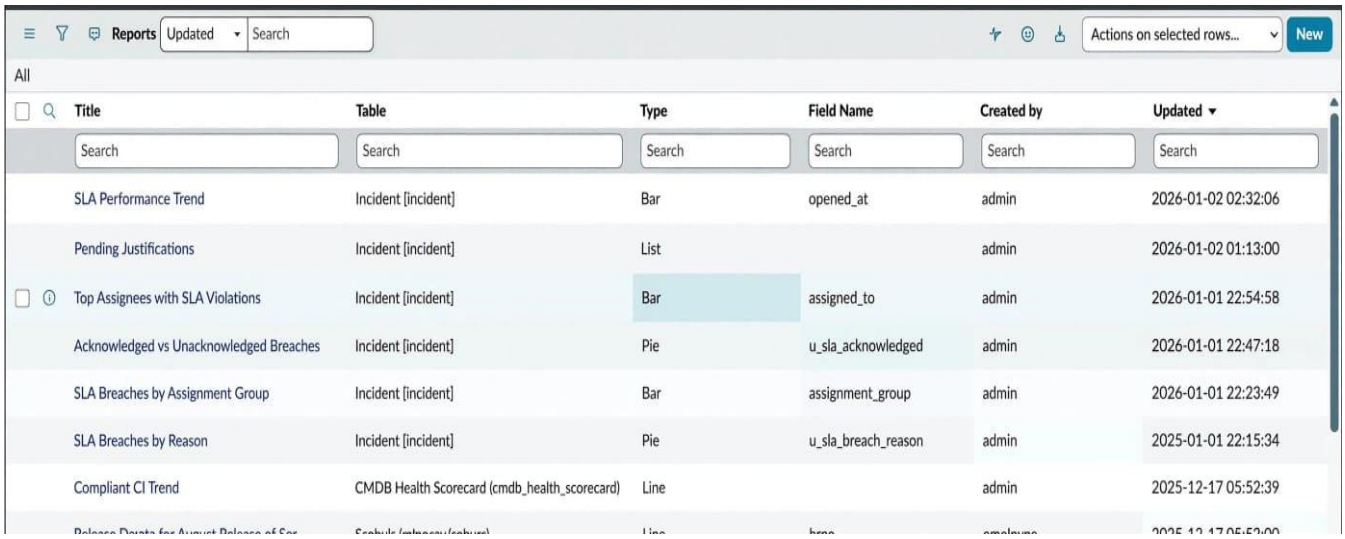


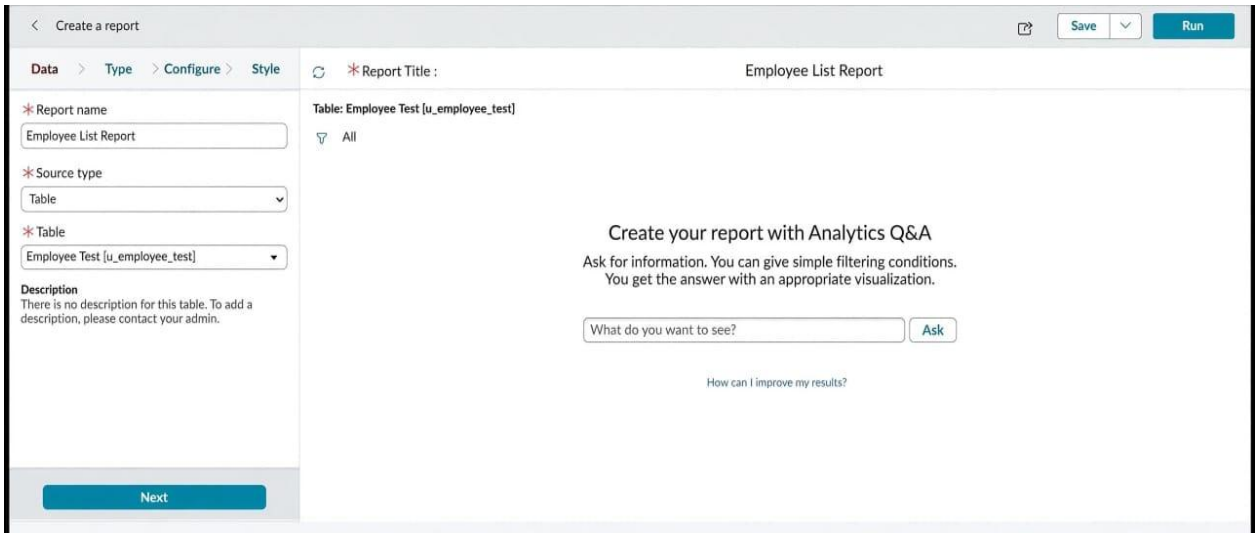
Report 3: Employee List Report.

Configuration

- Create a New Report name : **Employee List Report.**
- Table: **Employee Test.**
- Select Type: **List.**
- In Configure Select Columns:
    - **Employee ID.**
    - **Employee Name.**
    - **Email.**
    - **Department.**
    - **Location.**
- Now Select **next** and no required to choose anything in **Style**.
- Now Click **Run** to Execute the Report and Click **Save** to Save the Report.




Reports
Updated
Search
Actions on selected rows...
New
All
Title
Table
Type
Field Name
Created by
Updated
SLA Performance Trend
Incident [incident]
Bar
opened_at
admin
2026-01-02 02:32:06
Pending Justifications
Incident [incident]
List
admin
2026-01-02 01:13:00
Top Assignees with SLA Violations
Incident [incident]
Bar
assigned_to
admin
2026-01-01 22:54:58
Acknowledged vs Unacknowledged Breaches
Incident [incident]
Pie
u_sla_acknowledged
admin
2026-01-01 22:47:18
SLA Breaches by Assignment Group
Incident [incident]
Bar
assignment_group
admin
2026-01-01 22:23:49
SLA Breaches by Reason
Incident [incident]
Pie
u_sla_breach_reason
admin
2025-01-01 22:15:34
Compliant CI Trend
CMDB Health Scorecard (cmdb_health_scorecard)
Line
admin
2025-12-17 05:52:39


Create a report
Save
Run
Data
Type
Configure
Style
Report Title :
Employee List Report
Report name
Employee List Report
Source type
Table
Table
Employee Test [u_employee_test]
Description
There is no description for this table. To add a description, please contact your admin.
Table: Employee Test [u_employee_test]
All
Create your report with Analytics Q&A
Ask for information. You can give simple filtering conditions.
You get the answer with an appropriate visualization.
What do you want to see?
Ask
How can I improve my results?
Next

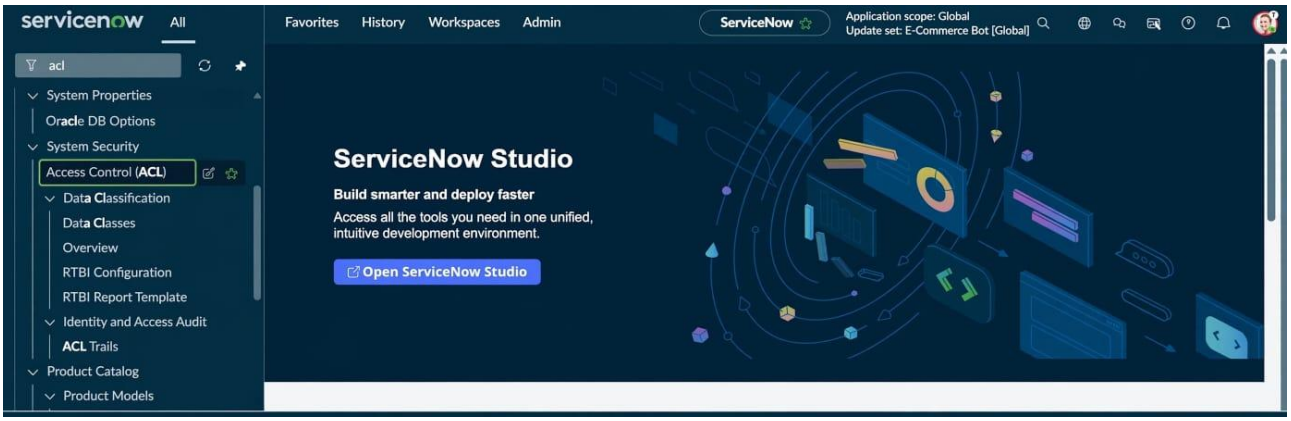
PHASE 9 : Add Reports to Dashboard

Create Dashboard

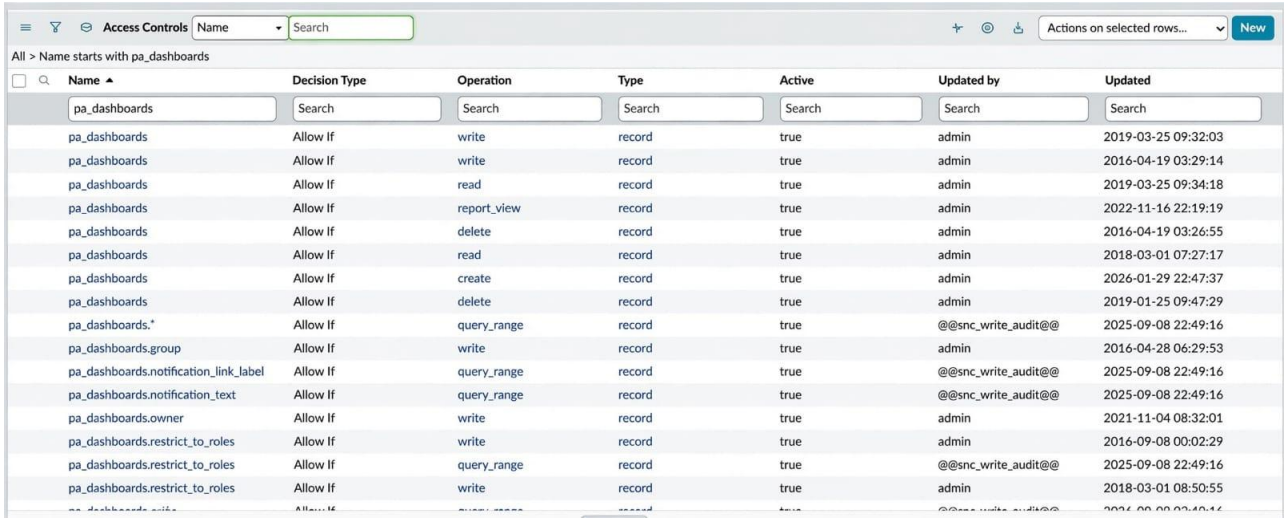
Navigation :

- Open All menus → Search for Dashboards.
- Under Self Service → Click Dashboards.
- On the top right corner Click Create Dashboards or /
- You can create Directly by using the backend name [PA_DASHBOARDS.FORM].
- Then Click New to Create new Dashboards.

- Only the backend name will open the form if the **ACL** has the **pa_dashboards** access to **Admin Overrides** True only.
- Then only we have access to the form.



- After Opening the ACL Search the pa_dashboards on Name Column.

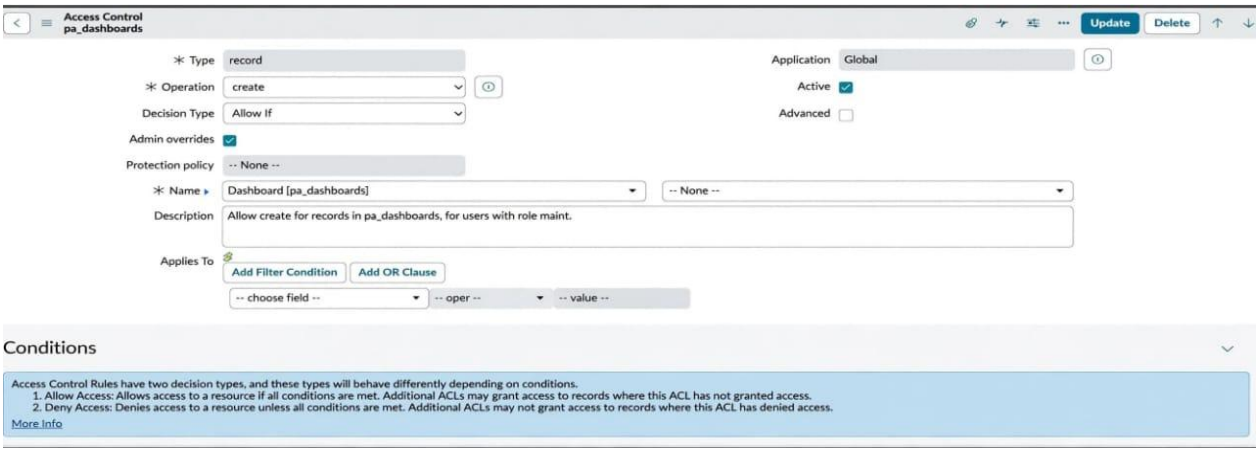


Access Controls | Name | Search | Actions on selected rows... | New

All > Name starts with pa_dashboards

Name	Decision Type	Operation	Type	Active	Updated by	Updated
pa_dashboards	Search	Search	Search	Search	Search	Search
pa_dashboards	Allow If	write	record	true	admin	2019-03-25 09:32:03
pa_dashboards	Allow If	write	record	true	admin	2016-04-19 03:29:14
pa_dashboards	Allow If	read	record	true	admin	2019-03-25 09:34:18
pa_dashboards	Allow If	report_view	record	true	admin	2022-11-16 22:19:19
pa_dashboards	Allow If	delete	record	true	admin	2016-04-19 03:26:55
pa_dashboards	Allow If	read	record	true	admin	2018-03-01 07:27:17
pa_dashboards	Allow If	create	record	true	admin	2026-01-29 22:47:37
pa_dashboards	Allow If	delete	record	true	admin	2019-01-25 09:47:29
pa_dashboards.*	Allow If	query_range	record	true	@@snc_write_audit@@	2025-09-08 22:49:16
pa_dashboards.group	Allow If	write	record	true	admin	2016-04-28 06:29:53
pa_dashboards.notification_link_label	Allow If	query_range	record	true	@@snc_write_audit@@	2025-09-08 22:49:16
pa_dashboards.notification_text	Allow If	query_range	record	true	@@snc_write_audit@@	2025-09-08 22:49:16
pa_dashboards.owner	Allow If	write	record	true	admin	2021-11-04 08:32:01
pa_dashboards.restrict_to_roles	Allow If	write	record	true	admin	2016-09-08 00:02:29
pa_dashboards.restrict_to_roles	Allow If	query_range	record	true	@@snc_write_audit@@	2025-09-08 22:49:16
pa_dashboards.restrict_to_roles	Allow If	write	record	true	admin	2018-03-01 08:50:55

- Now Open the **Create Operation** row.



Access Control
pa_dashboards

Update | Delete

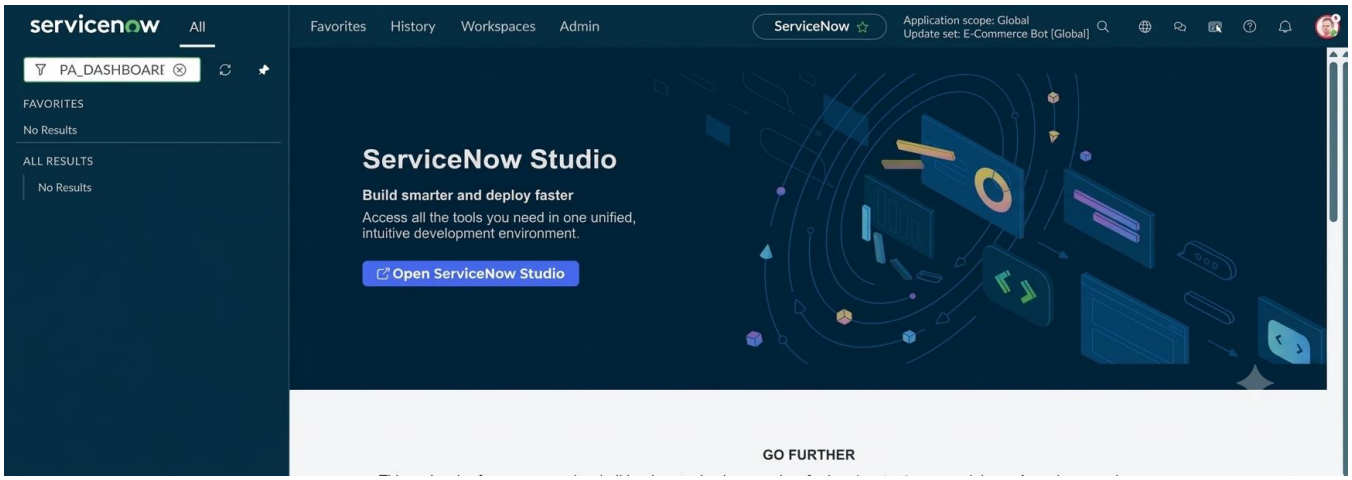
* Type: record
* Operation: create
Decision Type: Allow If
Admin overrides
Protection policy: -- None --
* Name: Dashboard [pa_dashboards] | -- None --
Description: Allow create for records in pa_dashboards, for users with role maint.
Applies To: Add Filter Condition | Add OR Clause
-- choose field -- | -- oper -- | -- value --

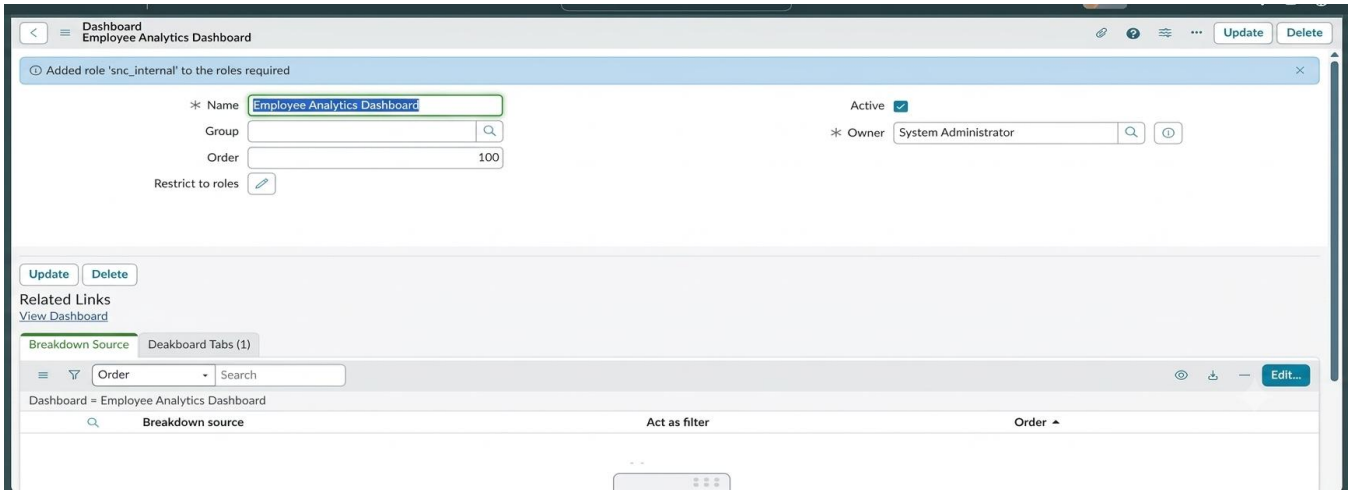
Application: Global
Active
Advanced

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
More Info

- Then Click Enable the True for Admin Overrides and Update the Form.
- Then the creation of dashboards will be possible by using the backend name.
- Now Open the **PA_DASHBOARDS.FORM** create dashboard.
- Dashboard name : **Employee Analytics Dashboards.**
- **Save** the Form.



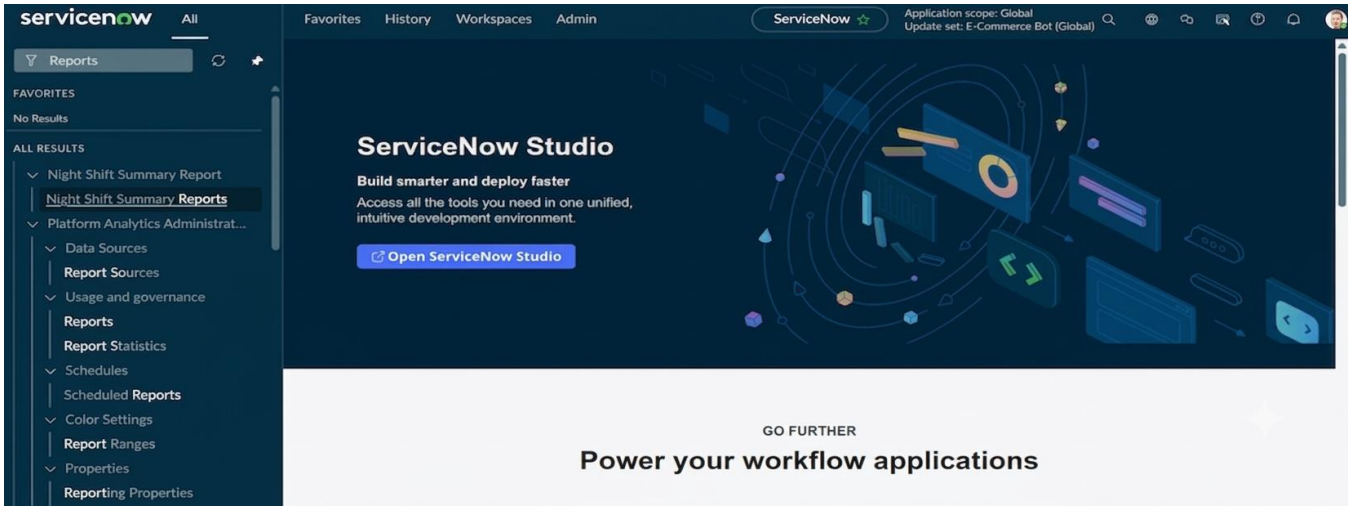


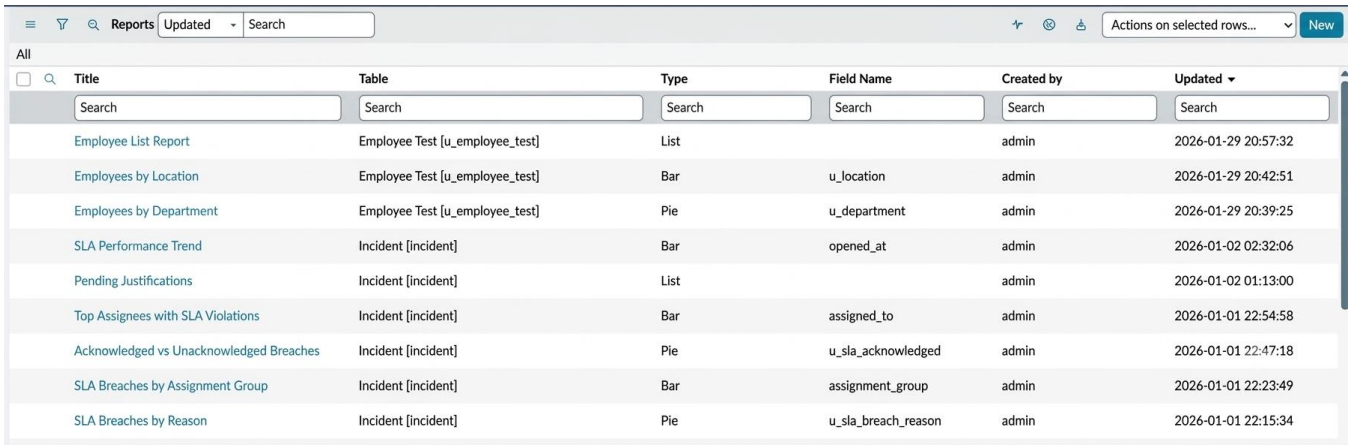
Add Reports to Dashboards.

Navigation Steps:

- Open the **Previously Created Reports.**
- On All menus → type Reports Under **Usage and Governance** select **reports**.
- Open the **Employees by Department**.
- Select the **Edit Report** and Click the **Share** button then the right top corner Select the **Add to Dashboards** Option.
- Then Choose the Created **dashboard name** and Click **Add** button.

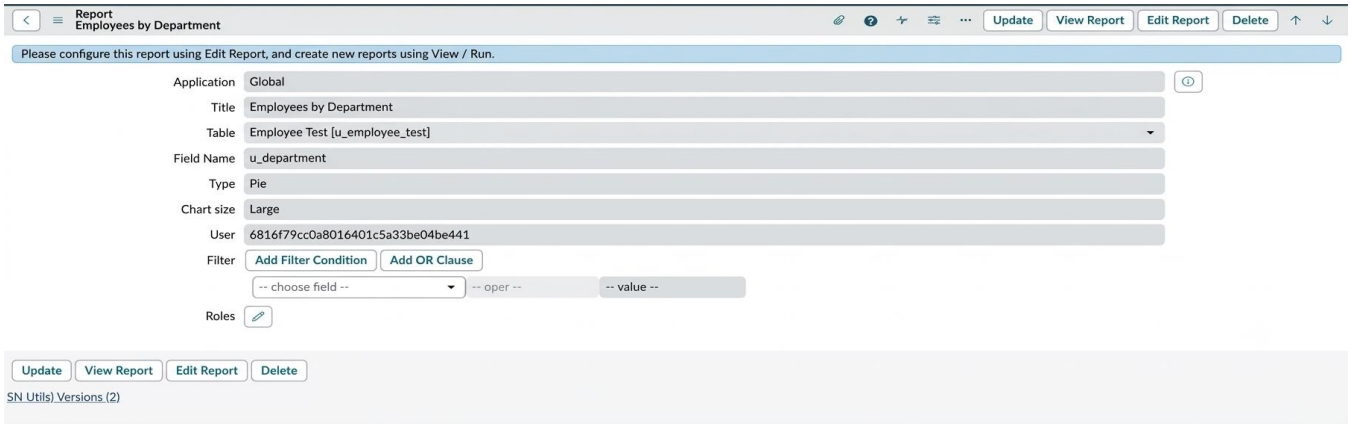
- Same Like that do the remaining **2 reports** in the same navigation how we added **Employees by Department.**
- The **Employees by Location** and **Employee List Report** these two reports are also added to the created dashboard.



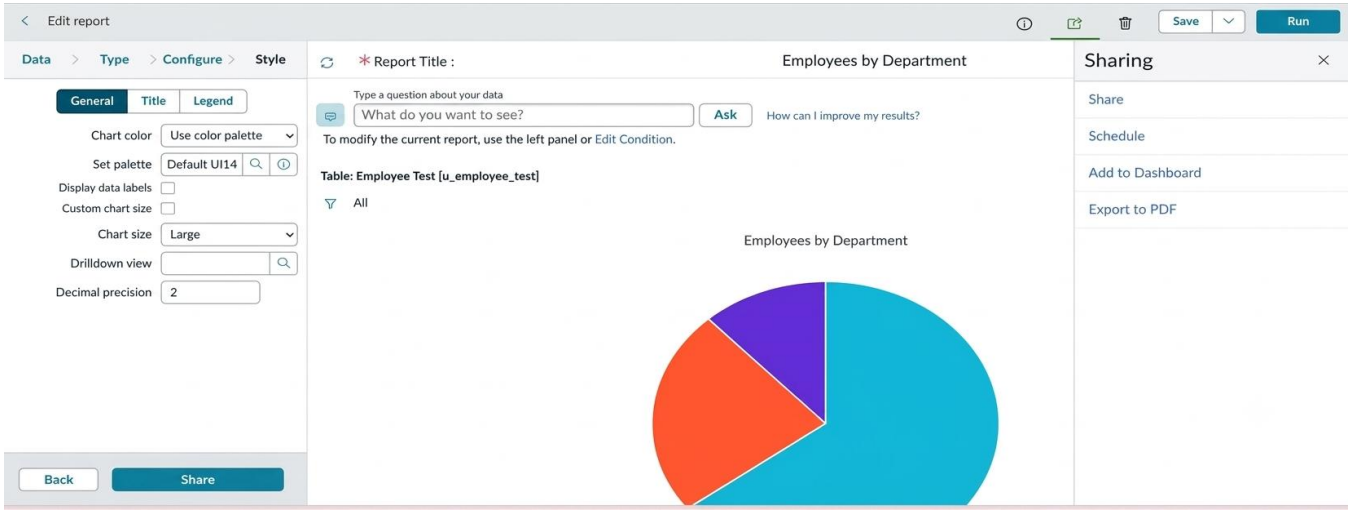


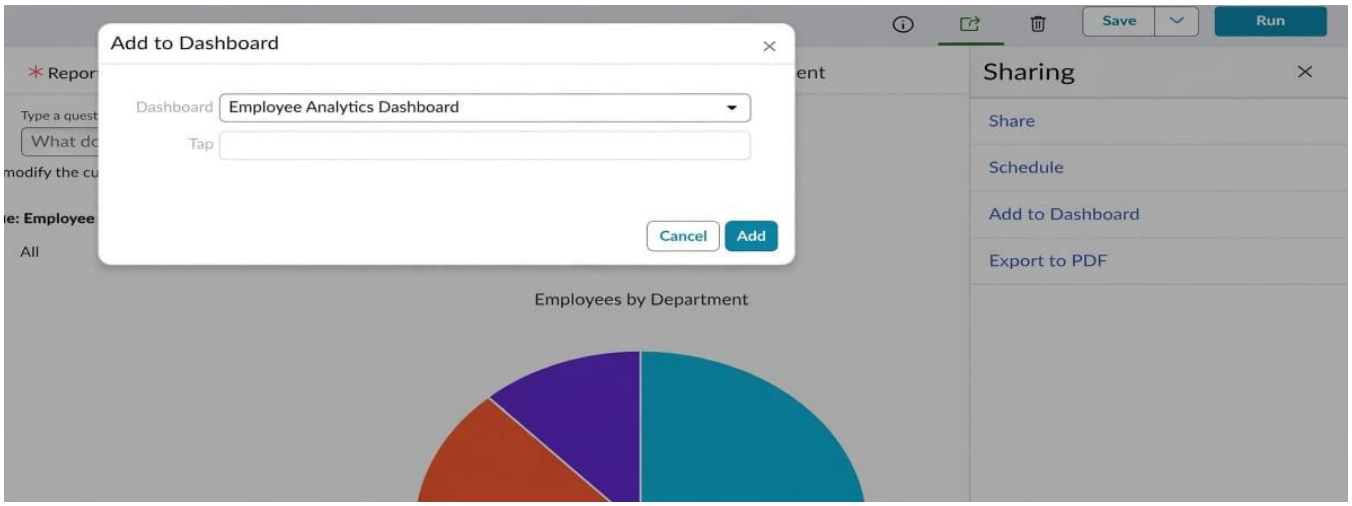
Title	Table	Type	Field Name	Created by	Updated
Employee List Report	Employee Test [u_employee_test]	List		admin	2026-01-29 20:57:32
Employees by Location	Employee Test [u_employee_test]	Bar	u_location	admin	2026-01-29 20:42:51
Employees by Department	Employee Test [u_employee_test]	Pie	u_department	admin	2026-01-29 20:39:25
SLA Performance Trend	Incident [incident]	Bar	opened_at	admin	2026-01-02 02:32:06
Pending Justifications	Incident [incident]	List		admin	2026-01-02 01:13:00
Top Assignees with SLA Violations	Incident [incident]	Bar	assigned_to	admin	2026-01-01 22:54:58
Acknowledged vs Unacknowledged Breaches	Incident [incident]	Pie	u_sla_acknowledged	admin	2026-01-01 22:47:18
SLA Breaches by Assignment Group	Incident [incident]	Bar	assignment_group	admin	2026-01-01 22:23:49
SLA Breaches by Reason	Incident [incident]	Pie	u_sla_breach_reason	admin	2026-01-01 22:15:34

- After Opening the Report Click the **View Report** Button on the right side of the Context menu.

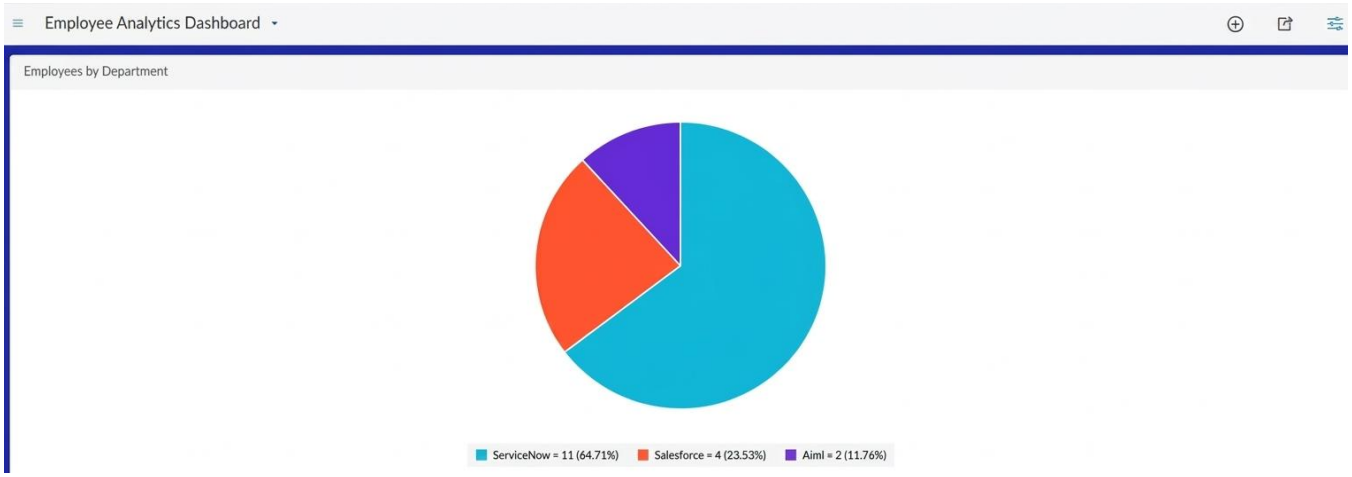


- After Clicking the View Report button, Now Click the **Share** button and then share options are visible on the right side.
- Now click the **Add to Dashboards**.





- Now Add the Dashboard that you created called **Employee Analytics Dashboards.**
- Then Click **Add** button.
- Now the **Report** is added to the **dashboards**.
- Now, Same in the process, add the remaining **two reports** to the **dashboard**.



- From the + icon also we add widgets that our reports from there.
- If we want we can also configure the background color.
- Here the three Reports are added successfully by varying **Department , Location and List Report.**
- We can also restrict the **Users,Groups and Roles** From Share Option.

Conclusion:

- This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency, and data integrity throughout the employee onboarding and update process.

- A key enhancement in this project was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system intelligently identifies existing employee records during every import. This prevents the creation of duplicate records and ensures that repeated imports update existing data instead of inserting redundant entries. As a result, the employee database remains clean, reliable, and consistent, which is critical in real-world enterprise environments where data may be imported multiple times from external systems.

- In addition to data integrity, the project was extended with Reports and Dashboards to provide meaningful insights and real-time visibility into employee data and import activities. Custom reports were created to track:
  - ✓ Total employee records.
  - ✓ Newly added employees.
  - ✓ Updated employee records.
  - ✓ Department-wise employee distribution.
  - ✓ Import success and error counts.

- These reports were then consolidated into a Dashboard, allowing administrators and HR teams to monitor employee data trends, import performance, and data quality from a single centralized view. Dashboards improved decision-making by presenting information visually through charts and graphs, reducing the need for manual analysis.

- Thus, this entire project including data import, transformation, and dashboard creation was successfully designed, implemented, and completed by us